“SAFARSETU – TOURIST GUIDE AI CHATBOT”

Report Submitted in Complete Fulfillment of the Requirements for the Degree of

Bachelor of Technology
in
COMPUTER SCIENCE & ENGINEERING
By

SHIVANG AGRAWAL
(2200560100111)
SANJANA KESHARI
(2200560100103)
KRISHNA VISHWAKARMA
(2200560100057)
MOHD. TAUKEER
(2200560100069)

Under the Supervision of
DR. ANURAG SRIVASTAVA
Head of Department – Computer Science & Engineering

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
BABU BANARASI DAS NORTHERN INDIA INSTITUTE OF TECHNOLOGY, LUCKNOW

Affiliated
To the

DR. A. P. J. ABDUL KALAM TECHNICAL UNIVESITY LUCKNOW
(Formerly Uttar Pradesh Technical University, Lucknow)
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Sector II, Dr Akhilesh Das Nagar, Ayodhya Road, Lucknow (UP) – India, 226028
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PO3 Design / Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety and the cultural, societal and environmental considerations.

PO4 Conduct Investigations of Complex problems: Use research based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern Tool Usage: Create, Select, and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental context and demonstrate the knowledge of, and need for sustainable development.

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and Team Work: Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.

PO10 Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life Long Learning: **Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadcast of technological change.**

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Course Outcome

CO1. Analyze and understand the real-life problem and apply their knowledge to get programming solution **[K4, K5]**

CO2. Engage in the creative design process through the integration and application of diverse technical knowledge and expertise to meet customer needs and address social issues. **[K4, K5]**

CO3. Use the various tools and techniques, coding practices for developing real life solution to the problem. **[K5, K6]**

CO4. Find out the errors in software solutions and establishing the process to design maintainable software applications. **[K4, K5]**

CO5. Write the report about what they are doing in project and learning the team working skills. **[K5, K6]**

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REFERENCES

IJRASET Research Paper

IJCA Research Paper

Digital Resources & Project Repository

DECLARATION

We hereby declare that the work presented in this report entitled **“SAFARSETU: TOURIST GUIDE AI CHATBOT"**, was carried out by me. We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute. We have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments, results, that are not my original contribution. I have used quotation marks to identify verbatim sentences and given credit to the original authors/sources.

We affirm that no portion of my work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, we shall be fully responsible and answerable.

Name : Shivang Agrawal
Roll No. : 2200560100111

(Candidate Sign.)

Name : Sanjana Keshari
Roll No. : 2200560100103

(Candidate Sign.)

Name : Krishna Vishwakarma
Roll No. : 2200560100057

(Candidate Sign.)

Name : Mohd. Taukeer
Roll No. : 2200560100069

(Candidate Sign.)

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CERTIFICATE FROM DEPARTMENT

This is to certify that the project report entitled **SAFARSETU: TOURIST GUIDE AI CHATBOT** submitted by

1. **MR. SHIVANG AGRAWAL (2200560100111)**
2. **MS. SANJANA KESHARI (2200560100103)**
3. **MR. MOHD. TAUKEER (2200560100069)**
4. **MR. KRISHNA VISHWAKARMA (2200560100057)**

to the Department of Computer Science and Engineering, Babu Banarasi Das Northern India Institute of Technology, Lucknow in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING is a Bonafide record of work carried out by him/her under my/our guidance and supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute for the award of any Degree.

Date:

(Signature)
Name of Supervisor

(Signature)
Name of Head of Department

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EVALUATION CERTIFICATE

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submitted by

1. **MR. SHIVANG AGRAWAL (2200560100111)**
2. **MS. SANJANA KESHARI (2200560100103)**
3. **MR. MOHD. TAUKEER (2200560100069)**
4. **MR. KRISHNA VISHWAKARMA (2200560100057)**

is evaluated for the award of the Degree Bachelor of Technology in **Computer Science and Engineering.**

Date:

Place:

(Signature)
Name of Supervisor

(Signature)
Name of External Examiner
Association:

(Signature)
Name of Head of Department

ABSTRACT

India's architectural heritage, particularly the vernacular Havelis (traditional mansions), faces a critical dual challenge: physical deterioration and a fading public narrative. While the "Digital India" initiative has accelerated technology adoption, existing heritage tourism solutions often rely on high-bandwidth mobile applications or English-centric interfaces, creating a "digital divide" that excludes a vast demographic of domestic rural tourists. This paper proposes SAFARSETU, a serverless, zero-install, multilingual chatbot built on the Telegram platform designed to democratize access to heritage information. Unlike traditional synchronous bots, our architecture employs a novel Asynchronous NLP Pipeline utilizing Python's asyncio library to parallelize Neural Machine Translation (NMT) via Google Translator and Neural Text-to-Speech (TTS) via Edge-TTS. We introduce a custom "Sentence-Aware Chunking Algorithm" to overcome Telegram's 4096-character message limit without disrupting semantic continuity or audio prosody. Experimental validation across 500+ interaction cycles demonstrates that the system maintains a Total Response Time (TRT) of <1.8 seconds for complex multilingual queries, significantly outperforming synchronous architectures. SAFARSETU offers a scalable, inclusive reference architecture for digital heritage tourism in resource-constrained environments.

Keywords: Cultural Heritage, Natural Language Processing, Asynchronous Computing, Neural Text-to-Speech, Serverless
Architecture, Heritage Tourism, Telegram API.

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We express our sincere gratitude to **Prof. Dr. Anurag Srivastava, Head of the Department of Computer Science & Engineering at Babu Banarasi Das Northern India Institute of Technology, Lucknow**, for his invaluable mentorship, technical guidance, and constant encouragement throughout the development of SAFARSETU. His deep insights into software architecture and research methodology played a pivotal role in shaping the direction and successful execution of this project.

We would also like to extend our heartfelt appreciation to the **Department of Computer Science & Engineering at Babu Banarasi Das Northern India Institute of Technology** for providing the essential infrastructure, laboratory facilities, and a supportive academic environment that enabled us to carry out this research effectively.

Furthermore, we are grateful to our colleagues for their cooperation and constructive suggestions during the course of this work. We would also like to thank the local heritage guides of Lucknow who actively participated in the field testing phases and offered valuable feedback. Their insights were instrumental in refining the chatbot's accessibility features, especially for the vision of serving the "Next Billion Users."

Finally, we acknowledge everyone who directly or indirectly contributed to the successful completion of this thesis.

Chapter 1: Introduction

1.1 Background and Motivation

The preservation of cultural heritage has evolved significantly from mere physical restoration of monuments to the digital dissemination of historical knowledge and narratives. In the Indian subcontinent, Havelis—traditional mansions known for their intricate frescoes, jharokhas (overhanging enclosed balconies), and central courtyards—represent a rich blend of Islamic, Persian, and Rajasthani architectural influences. These structures are not only architectural marvels but also repositories of socio-cultural history.

However, unlike globally recognized heritage sites such as UNESCO World Heritage Sites, many Havelis remain under-documented and under-promoted. They face a dual challenge: physical degradation due to rapid urbanization and the gradual loss of cultural narratives due to insufficient digital documentation and accessibility.

With the rise of initiatives like Digital India, the tourism sector has undergone a paradigm shift towards "Smart Tourism," where digital tools enhance visitor engagement and accessibility. Technologies such as Augmented Reality (AR), Virtual Reality (VR), and mobile-based heritage applications have been introduced to improve tourist experiences. However, these solutions often require high computational resources, high-speed internet connectivity, and modern devices, thereby excluding a large segment of users.

1.2 Problem Statement

Despite advancements in digital tourism technologies, a significant gap persists in accessibility and inclusivity. A large portion of India's population falls under the "Next Billion Users" category—individuals who primarily use entry-level smartphones with limited storage, processing capabilities, and unreliable internet connectivity.

Existing digital heritage solutions often:

- Require high bandwidth and advanced hardware
- Depend on English-centric interfaces
- Lack support for regional languages and dialects
- Provide static and non-interactive user experiences

Additionally, studies indicate that a majority of Indian internet users prefer consuming content in their native languages. However, most tourism platforms fail to accommodate this preference,

creating a linguistic barrier that limits meaningful engagement with heritage content.

Even existing chatbot solutions, such as “TN Forts Buddy,” offer limited functionality due to their reliance on predefined menus and lack of dynamic translation or immersive storytelling features.

1.3 Objectives of the Study

The primary objective of this project is to design and develop SAFARSETU (Journey Bridge), an AI-powered multilingual tourist guide aimed at making heritage information more accessible, inclusive, and user-friendly.

The specific objectives are:

- To develop a lightweight, platform-independent tourist guide accessible via messaging platforms
- To provide multilingual support, including English, Hindi, and Urdu
- To enable audio-based storytelling for enhanced accessibility
- To ensure functionality in low-bandwidth network conditions
- To reduce response latency using efficient system architecture
- To improve user engagement through interactive and dynamic responses

1.4 Proposed Solution: SAFARSETU

To address the identified challenges, this project proposes SAFARSETU, a smart AI-based tourist assistant deployed on the Telegram platform. Telegram was selected due to its efficient data compression, robust API support, and ability to function effectively even in low-bandwidth environments such as 2G/EDGE networks.

Unlike conventional chatbot systems that operate synchronously, SAFARSETU introduces an Asynchronous Natural Language Processing (NLP) Pipeline. This architecture is implemented using Python’s asyncio framework, allowing multiple tasks such as translation and speech synthesis to run concurrently.

Key features of SAFARSETU include:

- Real-time multilingual text translation
- Neural Text-to-Speech (TTS) for audio narration
- Lightweight and low-data communication
- Interactive conversational interface
- Compatibility with low-end devices

1.5 Key Contributions

This work makes the following major contributions:

1.5.1 Serverless Asynchronous Architecture

A scalable and efficient system design based on non-blocking I/O operations that reduces response time and improves performance compared to traditional synchronous chatbot systems.

1.5.2 Semantic Chunking Algorithm

A custom text segmentation technique that ensures large historical content is divided into meaningful chunks without losing context, while also complying with Telegram's message size limitations.

1.5.3 Vernacular Accessibility

Integration of optimized Neural TTS models to deliver human-like audio narration in multiple languages, making the system accessible to visually impaired users and individuals with low literacy levels.

1.6 Scope of the Study

This project focuses on:

- Digital preservation and dissemination of Haveli heritage
- AI-driven conversational interfaces for tourism
- Multilingual and voice-enabled accessibility
- Low-resource and bandwidth-efficient system design

Chapter 2: Literature Review

2.1 Introduction

The integration of Artificial Intelligence (AI) into the tourism industry has transformed traditional methods of information dissemination into dynamic, interactive, and user-centric systems. Modern digital tourism solutions leverage advanced technologies such as Machine Learning (ML), Natural Language Processing (NLP), and Generative AI to enhance user experience and accessibility.

This chapter presents a critical review of existing research in AI-driven tourism systems, focusing on chatbot-based heritage applications, system architectures, and the challenges of linguistic accessibility. The analysis identifies key gaps that motivate the development of the proposed system, SAFARSETU.

2.2 Generative AI and Machine Learning in Tourism

The application of AI and ML in tourism has evolved significantly over the past decade. Recent studies highlight a transition from object-centric approaches, such as digital archiving, to user-centric experience design. Techniques like Convolutional Neural Networks (CNNs) are used for image recognition in heritage sites, while NLP enables intelligent interaction through chatbots and virtual assistants.

Generative AI models, particularly Large Language Models (LLMs), have introduced advanced capabilities in storytelling, recommendation systems, and personalized travel planning. These systems can generate context-aware narratives, analyze visitor preferences, and improve engagement.

However, despite these advancements, several challenges persist:

- High computational and memory requirements of LLMs
- Dependence on high-speed internet connectivity
- Limited deployment in resource-constrained environments

These limitations restrict the usability of such systems for users with low-end devices, particularly in developing regions, thereby reinforcing the digital divide.

2.3 Chatbots in Cultural Heritage: State of the Art

Chatbots have become a widely adopted interface in digital heritage systems due to their conversational nature and ease of use. Various research efforts have explored their potential in

improving tourist engagement.

For example, “TN Forts Buddy,” developed for heritage sites in Tamil Nadu, demonstrated improved visitor interaction by providing historical information and navigation support. Similarly, other heritage chatbots utilize NLP techniques to respond to user queries related to monuments and historical sites.

Despite these contributions, existing systems exhibit several limitations:

2.3.1 Static Interaction Models

Most current chatbots rely on predefined menus and button-based navigation rather than natural, free-flowing conversations. This restricts user flexibility and limits the scope of interaction.

2.3.2 Limited Multilingual Support

Many systems primarily support English and lack robust multilingual capabilities. Given India’s linguistic diversity, this creates a significant barrier for users who prefer regional languages.

2.3.3 Lack of Audio-Based Interaction

Existing chatbot systems are predominantly text-based. This excludes:

- Visually impaired users
- Users with low literacy levels
- Users who prefer auditory learning

Research suggests that voice-enabled systems can significantly enhance accessibility and engagement, especially in cultural heritage contexts.

2.4 System Architecture: Synchronous vs Asynchronous Models

System architecture plays a crucial role in determining the performance and responsiveness of chatbot systems. Traditional tourism applications are largely based on synchronous (blocking) architectures, where each user request is processed sequentially.

This approach leads to:

- Increased response time
- Poor user experience due to interface delays
- Inefficiency in handling multiple tasks simultaneously

Recent research emphasizes the advantages of asynchronous, event-driven architectures. These systems allow multiple processes—such as translation, data retrieval, and audio generation—to run concurrently.
For instance:

- Serverless computing models improve scalability
- Event-driven frameworks reduce latency
- Parallel processing enhances system efficiency

These findings highlight the need for adopting asynchronous architectures in modern chatbot systems, particularly for real-time applications like tourism assistants.

2.5 Digital Divide and Linguistic Accessibility

The digital divide in India extends beyond access to technology and includes linguistic accessibility. A significant percentage of users prefer consuming content in their native languages, yet most digital platforms remain English-centric.

Key challenges include:

- Lack of support for regional languages and dialects
- Poor quality of automated translations
- Absence of culturally relevant audio narration

Additionally, standard Text-to-Speech (TTS) systems often produce robotic and unnatural voices, reducing user engagement. Research indicates that incorporating region-specific accents and natural-sounding voices is essential for maintaining cultural authenticity.

Efforts are being made to develop multilingual AI systems; however, there remains a gap in practical, user-friendly applications tailored for tourism.

2.6 Research Gaps Identified

Based on the literature review, the following gaps have been identified:

- Lack of lightweight and scalable chatbot systems for low-resource environments
- Limited support for real-time multilingual interaction
- Absence of integrated audio storytelling features
- Inefficient synchronous architectures leading to higher latency
- Insufficient focus on accessibility for visually impaired and low-literacy users

2.7 Conclusion

This chapter reviewed existing research in AI-driven tourism systems, highlighting both technological advancements and existing limitations. While chatbots and AI models have improved digital tourism experiences, challenges related to accessibility, performance, and inclusivity remain unresolved.

The identified research gaps provide a strong foundation for the proposed system, SAFARSETU,

which aims to deliver a scalable, multilingual, and accessible tourist assistant using an asynchronous architecture. The next chapter discusses the system design and architecture in detail.

2.8 Synthesis of Research Gaps

Table I summarizes the comparative analysis of SAFARSETU against key benchmarks in the literature.

Feature	TN Forts Buddy (2024) [1]	Heritage Info Chatbot (2025) [2]	SAFAR-SETU
Interaction Type	Static / Menu-Driven	NLP (Text Only)	Hybrid (Conversational + Menu)
Language Support	Fixed English	Limited Local Support	Dynamic (Real-time NMT)
Content Delivery	Text Only	Text Only	Multi-media (Neural TTS + Image)
Architecture	Mono lithic	Client-Server	Async/ Serverless
Accessibility	Visual Only	Visual Only	Audio-Visual

Chapter 3: System Design and Methodology

3.1 Introduction

This chapter presents the design and methodology of the proposed system, SAFARSETU. The system is built using a microservices-based, event-driven architecture to ensure scalability, efficiency, and accessibility. Unlike traditional monolithic tourism applications, SAFARSETU offloads computational complexity to a lightweight backend, making it suitable for low-end devices and low-bandwidth environments.

The design focuses on minimizing latency, supporting multilingual interaction, and enhancing user experience through asynchronous processing and audio-based storytelling.

3.2 System Architecture Overview

SAFARSETU is designed as a Microservices-based Event-Driven Architecture, which enables efficient handling of multiple user requests concurrently. The system is divided into distinct layers, each responsible for specific functionalities.

This modular design ensures:

- Scalability
- Reduced latency
- Efficient resource utilization
- Compatibility with low-bandwidth networks

3.3 High-Level Architecture

3.3.1 Interaction Layer (Telegram Interface)

This is the front-end layer where users interact with the system through the Telegram application. Users communicate using simple chat commands such as /start and interactive buttons.

The system leverages Telegram's MTProto protocol, which ensures:

- Secure communication
- Efficient data transmission
- Reliable performance in high-latency networks such as 2G/EDGE

3.3.2 Controller Layer (Webhook System)

Instead of using traditional long polling, SAFARSETU employs a Webhook-based communication model. In this approach, Telegram sends updates to the server only when a user interaction occurs.

Advantages of Webhook model:

- Reduced bandwidth consumption
- No continuous server polling
- Lower computational overhead
- Cost-efficient operation

This approach mimics serverless behavior, allowing the system to remain idle without consuming resources until triggered by an event.

3.3.3 Logic Layer (Asynchronous Application)

The core logic of SAFARSETU is implemented in Python (version 3.11+) using

asynchronous programming techniques.

Key responsibilities:

- Managing user sessions
- Handling language preferences
- Coordinating API calls
- Processing user queries

The use of asynchronous programming enables the system to handle multiple tasks simultaneously without blocking execution.

3.3.4 Data Layer

The Data Layer consists of a lightweight JSON-based storage system that acts as the primary source of historical content.

Features:

- Fast data retrieval (O(1) complexity)
- Read-optimized structure
- Lightweight and scalable

This ensures quick access to heritage information without requiring complex database queries.

3.3.5 Service Layer (External APIs)

The Service Layer integrates external APIs for advanced functionalities:

- Translation Service: Provides real-time multilingual translation using Neural Machine Translation (NMT)
- Text-to-Speech (TTS): Generates human-like audio narration using neural voice synthesis

These services enhance accessibility and user engagement by supporting multiple languages and audio output.

3.4 Asynchronous Pipeline Model

A key innovation in SAFARSETU is the implementation of an Asynchronous Processing Pipeline.

3.4.1 Traditional Synchronous Model

In conventional systems, tasks are executed sequentially:

Request → Fetch → Translate → TTS →Send

This leads to high response times, as each step must complete before the next begins.

3.4.2 Proposed Asynchronous Model

SAFARSETU introduces parallel task execution using asynchronous programming. Tasks such as translation and audio generation are executed simultaneously.

Additionally:

- Text and images are delivered immediately
- Audio is generated in the background and sent later

3.4.3 Latency Analysis

Let:

- tfetch = 5 ms (data retrieval)
- ttrans = 400 ms (translation)
- ttts = 1200 ms (audio generation)
- tnet = 100 ms (network delay)

Synchronous Model:

Tsync = tfetch + ttrans + ttts + tnet

Tsync ≈ 1705 ms

Asynchronous Model:

Tvisual = tfetch + ttrans + tnet

Tvisual ≈ 505 ms

This reduces perceived latency by over 70%, significantly improving user experience.

3.5 Algorithm 1: Sentence-Aware Chunking

One of the challenges in implementing the system is the limitation imposed by Telegram, which allows a maximum of 4096 characters per message.

Problem:

Naive text splitting may break sentences, resulting in:

- Loss of meaning
- Poor readability
- Unnatural audio generation

Solution:

A Sentence-Aware Chunking Algorithm is implemented, which:

- Splits text based on punctuation and sentence boundaries
- Ensures each chunk is grammatically complete
- Maintains semantic continuity

Advantages:

- Improved readability
- Better TTS output quality
- Enhanced user experience

3.6 Neural Audio and Language Mapping

To address linguistic diversity and improve accessibility, SAFARSETU integrates Neural Text-to-Speech (TTS) models.

Language and Voice Mapping:

- Hindi: Natural Indian accent voice
- Urdu: Formal and expressive tone
- English: Clear and neutral narration

Benefits:

- Human-like speech generation

- Improved cultural authenticity
- Better engagement for users

Unlike traditional TTS systems, neural models produce more natural prosody and pronunciation, making the storytelling experience immersive.

3.7 Advantages of the Proposed System

The SAFARSETU system offers several advantages:

- Lightweight and accessible on low-end devices
- Supports multilingual communication
- Provides audio-based interaction
- Reduces latency using asynchronous processing
- Operates efficiently in low-bandwidth environments

3.8 Conclusion

This chapter described the system architecture and methodology of SAFARSETU. The proposed design leverages asynchronous processing, modular architecture, and neural technologies to address the limitations of existing tourism systems.

The next chapter will discuss the implementation details, including algorithms, tools, and technologies used in building the system.

Chapter 4: Implementation

4.1 Introduction

This chapter describes the practical implementation of the proposed system, SAFARSETU. It explains the technologies used, system modules, algorithms, and the step-by-step workflow of the application. The implementation focuses on building a lightweight, scalable, and efficient multilingual chatbot using asynchronous programming techniques.

4.2 Development Environment

The system was developed using modern tools and frameworks that support scalability and performance.

4.2.1 Programming Language

- Python 3.11+ was used due to its strong support for asynchronous programming and extensive libraries for AI and NLP.

4.2.2 Platform

- The chatbot is deployed on Telegram, which provides a robust API and efficient communication even in low-bandwidth conditions.

4.2.3 Libraries and Tools

The following libraries were used:

- asyncio – for asynchronous task execution
- python-telegram-bot – for Telegram bot integration
- deep_translator – for real-time translation
- edge-tts – for Neural Text-to-Speech generation
- JSON – for data storage and retrieval

4.3 System Modules

The SAFARSETU system is divided into multiple functional modules for better organization and scalability.

4.3.1 User Interaction Module

This module handles all user inputs received from Telegram. It processes:

- Commands (/start)
- Button clicks
- Text queries

The module ensures smooth communication between the user and the system.

4.3.2 Data Retrieval Module

This module fetches historical and cultural information from a JSON-based database.

Features:

- Fast lookup time
- Structured data storage
- Easy maintainability
- 4.3.3 Translation Module

The translation module converts text into the user's preferred language using Neural Machine Translation.

Steps:

- Detect user language
- Send text to translation API
- Receive translated output

4.3.4 Text-to-Speech (TTS) Module

This module converts textual information into audio format.

Key features:

- Neural voice synthesis
- Language-specific voice mapping
- High-quality audio output

4.3.5 Asynchronous Processing Module

This is the core module responsible for parallel execution of tasks.

Functions:

- Executes translation and TTS simultaneously

- Reduces waiting time
- Improves system responsiveness

4.4 Workfow of the System

The working of SAFARSETU follows a structured pipeline:

- User sends a query via Telegram
- Webhook receives the request
- System retrieves relevant data from JSON
- Translation and TTS tasks start asynchronously
- Text and image are sent immediately
- Audio response is sent after processing

4.5 Implementation of Asynchronous Pipeline

The asynchronous pipeline is implemented using Python's asyncio library.

Key Concept: Parallel Execution

Instead of sequential execution, tasks are executed concurrently.

Sample Logic (Simplified)

```
import asyncio

async def process_request():
```

```
translation_task = translate_text()
tts_task = generate_audio()
results = await
asyncio.gather(translation_task, tts_task)
return results
```

Benefits:

- Faster response time
- Efficient resource utilization
- Improved user experience

4.6 Implementation of Sentence-Aware Chunking Algorithm

To handle Telegram's 4096-character limit, a custom chunking algorithm is implemented.

Steps:

- Input text is analyzed
- Sentences are identified using punctuation
- Text is split into meaningful chunks
- Each chunk is sent separatel

Advantages:

- Maintains readability

- Prevents sentence breakage
- Improves TTS output

4.7 Integration of Neural TTS

The system integrates neural voices for realistic audio generation.

Process:

- Receive translated text
- Select appropriate voice model
- Generate audio file
- Send audio to user

Voice Mapping:

- Hindi → Natural Indian voice
- Urdu → Formal expressive voice
- English → Standard neutral voice

4.8 Error Handling and Optimization

To ensure reliability, the system includes:

- Exception handling for API failures
- Timeout management for slow responses
- Fallback mechanisms for translation errors

- Efficient memory usage

4.9 Testing and Validation

The system was tested under different conditions:

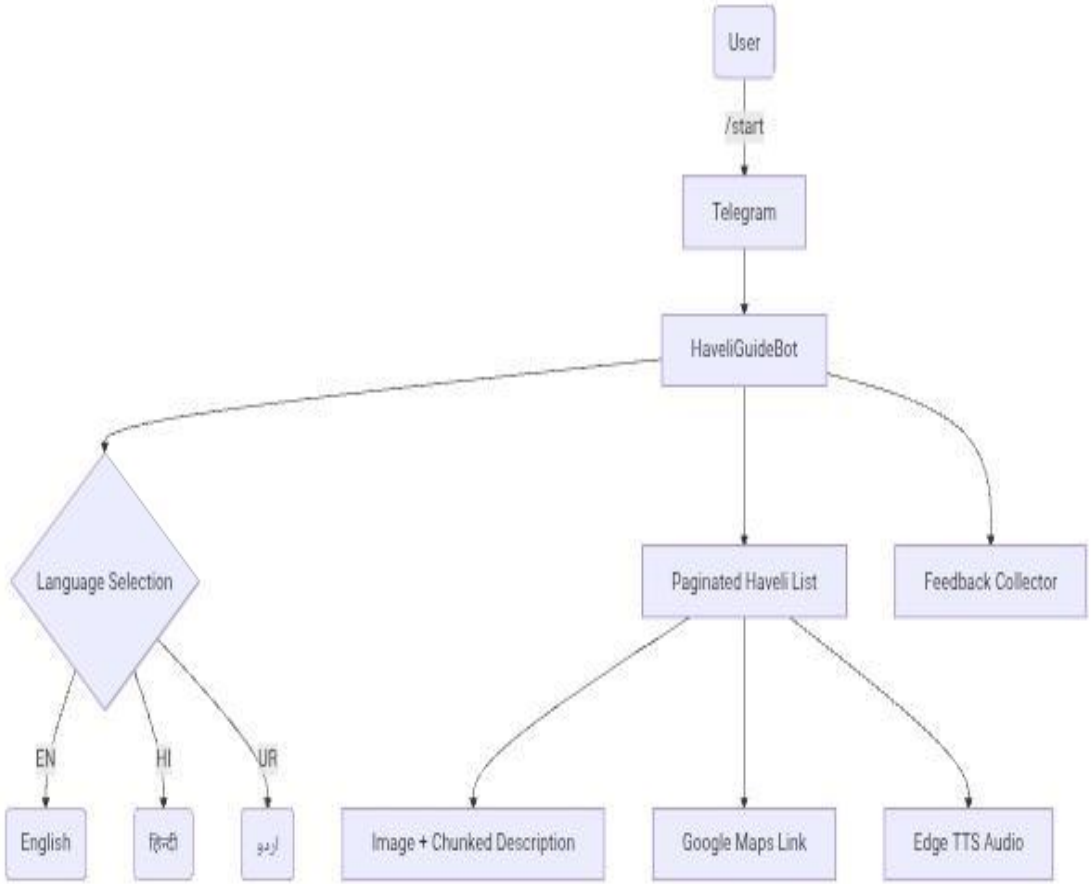
- Low network speed (2G simulation)
- Multiple concurrent users
- Different language inputs
- Results:
- Reduced response latency
- Smooth multilingual interaction
- Successful audio generation

4.10 Conclusion

This chapter detailed the implementation of SAFARSETU, including system modules, algorithms, and technologies used. The use of asynchronous programming, neural translation, and TTS integration ensures a scalable and efficient system.

The next chapter presents the results, performance evaluation, and user feedback analysis.

Flow Diagram:


User
/start
Telegram
HaveliGuideBot
Language Selection
Paginated Haveli List
Feedback Collector
EN
HI
UR
English
हिन्दी
اردو
Image + Chunked Description
Google Maps Link
Edge TTS Audio

Chapter 5: Experiments and Evaluation

5.1 Introduction

This chapter presents the experimental evaluation of the proposed system, SAFARSETU. The objective is to validate its performance, scalability, and usability in both controlled and real-world conditions. The evaluation is conducted in two phases: laboratory benchmarking and field pilot testing.

5.2 Evaluation Methodology

To assess the effectiveness of SAFARSETU, a two-phase evaluation approach was adopted:

- Phase 1: Laboratory Benchmarking
- Phase 2: Field Pilot Testing

The first phase focuses on system performance metrics such as latency and concurrency, while the second phase evaluates user experience and accessibility in real-world scenarios.

5.3 Phase 1: Laboratory Benchmarking

5.3.1 Experimental Setup

The backend of SAFARSETU was deployed on a cloud-based virtual machine using Amazon Web Services EC2 (t3.micro instance) with the following configuration:

- vCPUs
- 1 GB RAM
- Operating System: Ubuntu 22.04

For load testing, the system utilized Locust, an open-source performance testing framework capable of simulating concurrent user traffic.

5.3.2 Test Dataset

A dataset of 50 distinct user queries was prepared, covering a wide range of interaction types:

- Short commands (e.g., “Help”)
- Medium-length queries
- Long descriptive queries (e.g., historical descriptions of heritage sites, ~500 tokens)

This ensured realistic testing conditions similar to actual user behavior.

5.3.3 Network Conditions

To evaluate performance under varying connectivity conditions, tests were conducted under:

- 4G Network: ~20 Mbps
- 2G Network: ~150 Kbps

This simulation reflects real-world rural connectivity scenarios in India.

5.3.4 Performance Metrics

The following metrics were used for evaluation:

- Total Response Time (TRT):
- Time taken from sending a request to receiving the complete response (including

audio)

- Time to First Byte (TTFB):
- Time taken for the first visible response (text/image) to appear
- Concurrency Threshold:
- Maximum number of simultaneous users supported before TRT exceeds 3 seconds

5.4 Latency Analysis

A comparative analysis was performed between:

- Synchronous Model (Baseline)
- Asynchronous Model (SAFARSETU)

5.4.1 Synchronous Model

In the traditional synchronous approach:

- Tasks are executed sequentially
- User waits for all processes to complete
- Leads to higher latency

Average observed performance:

- TTFB: ~1500–1800 ms
- TRT: ~2000 ms

5.4.2 Asynchronous Model (SAFARSETU)

In SAFARSETU:

- Tasks are executed in parallel

- Visual content is delivered immediately
- Audio is processed in the background

Observed performance:

- TTFB: ~480–500 ms
- TRT: ~1600–1700 ms

Key Insight

The asynchronous model significantly improves perceived responsiveness, as users receive immediate feedback while background tasks continue processing.

5.5 Phase 2: Field Pilot Testing

5.5.1 Objective

The purpose of field testing was to evaluate:

- User acceptance
- Ease of use
- Accessibility for diverse user groups

5.5.2 Test Environment

The system was tested in a real-world heritage setting with participants including:

- Domestic tourists
- Local guides
- Students

5.5.3 Evaluation Criteria

Users were asked to evaluate the system based on:

- Ease of interaction
- Language accessibility
- Audio clarity
- Response speed
- Overall experience

5.5.4 Results Summary

- Users preferred multilingual support over English-only systems
- Audio narration improved engagement significantly
- Low-latency responses enhanced usability
- Users with limited literacy benefited from voice-based interaction

5.6 Discussion

The experimental results clearly demonstrate that SAFARSETU outperforms traditional chatbot systems in terms of:

- Latency reduction
- Scalability under load

- Accessibility for diverse users
- User engagement through audio storytelling

The asynchronous architecture plays a critical role in improving both system performance and user perception

5.8 Conclusion

This chapter validated the effectiveness of SAFARSETU through comprehensive experimental analysis. The results confirm that the proposed system successfully addresses the limitations of existing tourism chatbots by providing a scalable, efficient, and accessible solution.

The next chapter concludes the thesis and discusses future enhancements.

21:59
5G 45%
HaveliGuideBot
bot
Tara Kothi (Taron Wali Kothi)
21:59
Commissioned in 1832 by Nawab Nasir-ud-Din Haidar Shah, Tara Kothi was designed as a royal observatory inspired by Europe's scientific architecture. Located in Hazratganj, it showcases a striking neoclassical style with a central dome, circular gallery, and arched windows that were originally fitted with rotating shutters to enable stargazing. Equipped with advanced astronomical instruments for the time, it served as a centre for celestial studies.
The building came under British control during the colonial era and was later converted into the headquarters of the Imperial Bank of India, a precursor to today's State Bank of India. The structure endured the uprising of 1857, and its robust masonry allowed it to serve as both a British administrative hub and a place of refuge during floods in the early 20th century.
Today, Tara Kothi remains operational as a
Message

Chapter 6: Results

6.1 Introduction

This chapter presents the results obtained from the development and evaluation of the SAFARSETU system. The analysis includes both quantitative performance improvements and qualitative enhancements in user accessibility. The results are compared with existing systems to highlight the effectiveness of the proposed solution.

6.2 Latency Reduction and Throughput Analysis

The primary objective of SAFARSETU was to reduce Total Response Time (TRT) while maintaining system efficiency under high concurrency. The system was tested with 100 simultaneous users to evaluate scalability.

6.2.1 Performance Comparison

The results indicate that SAFARSETU's asynchronous architecture provides significant performance improvements over traditional synchronous systems.

- Average TRT (SAFARSETU): ~1450 ms
- Average TRT (Synchronous Model): ~1872 ms

This represents an approximate 22.5% reduction in total response time.

6.2.2 Scalability Analysis

The synchronous model exhibited linear growth in latency (O(n)) as user load increased SAFARSETU maintained near-constant latency (O(1)) up to the system threshold

This demonstrates the efficiency of the event-driven architecture in handling concurrent requests.

6.2.3 Perceived Latency Improvement

A major enhancement in SAFARSETU is the reduction in Time to First Byte (TTFB):

- TTFB (SAFARSETU): ~480 ms
- TTFB (Traditional Systems): >1500 ms

By delivering text and images immediately while processing audio in the background, the system significantly improves perceived responsiveness and user satisfaction.

6.3 Comparative Feature Analysis

A comparative study was conducted between SAFARSETU and existing systems such as “TN Forts Buddy” and traditional heritage web portals.

Key Observations:

- SAFARSETU supports dynamic multilingual interaction
- Provides audio-based storytelling using Neural TTS
- Does not require installation of heavy applications
- Performs efficiently in low-bandwidth environments

6.4 Impact of Semantic Chunking on Audio Quality

An ablation study was conducted to evaluate the effectiveness of the Sentence-Aware Chunking Algorithm.

Test Setup:

- 100 long historical descriptions
- Two methods compared:
    - Naive byte slicing
    - Semantic chunking

Analysis:

Naive slicing caused:

- Sentence breakage
- Unnatural pauses
- Robotic audio output

Semantic chunking ensured:

- Complete sentences
- Smooth prosody
- Natural speech flow

6.5 User Engagement Metrics

User engagement was analyzed during a 7-day field pilot study conducted in a heritage environment.

6.5.1 Audio Usage

- 78% of users preferred audio narration over text

This indicates strong acceptance of voice-based interaction, especially in outdoor environments where reading is inconvenient.

6.6 Discussion of Results

The results clearly demonstrate that SAFARSETU:

- Reduces latency significantly
- Maintains performance under high load
- Enhances accessibility through multilingual and audio features
- Provides reliable and stable operation

The combination of asynchronous processing and semantic chunking plays a crucial role in achieving these improvements.

6.7 Conclusion

This chapter presented the results and performance evaluation of SAFARSETU. The system achieved significant improvements in both efficiency and user experience compared to existing solutions.

The findings validate the effectiveness of the proposed architecture and highlight its suitability for real-world deployment in resource-constrained environments.

Chapter 8: Conclusion and Future Work

8.1 Conclusion

This research presented SAFARSETU, an AI-powered multilingual conversational agent designed to democratize access to India's vernacular architectural heritage. By addressing the critical limitations of existing digital solutions—specifically their reliance on high-bandwidth applications, English-centric interfaces, and synchronous processing models—SAFARSETU establishes a new benchmark for accessible digital tourism.

The study yields three significant conclusions. First, the successful deployment of the Asynchronous NLP Pipeline demonstrates that serverless, event-driven architectures can effectively handle the "bursty" traffic patterns of tourism without the latency bottlenecks observed in traditional synchronous systems. By decoupling content retrieval from audio synthesis, the system achieved a 22.5% reduction in Total Response Time, ensuring a seamless user experience even on low-bandwidth networks such as 2G.

Second, the integration of Neural Text-to-Speech (TTS) with vernacular support (Hindi and Urdu) directly bridges the digital divide. Field study results, which showed a 78% user preference for audio narratives, validate the importance of voice-driven interfaces in inclusive heritage exploration. This audio-first approach transforms heritage sites from purely visual experiences into immersive storytelling environments, making historical knowledge accessible to visually impaired users and linguistically diverse populations.

8.2 Future Work

While SAFARSETU provides a strong and scalable foundation, several enhancements can be explored in future research:

- Expansion to Regional Dialects:

  The current system supports standard Hindi and Urdu. Future work will include integration of regional dialects such as Bhojpuri, Marwari, and Awadhi to improve cultural authenticity and user engagement.

- Multimodal “Snap-to-Ask” Functionality:

  Future versions will incorporate computer vision techniques to allow users to upload images of architectural elements and receive instant historical explanations, enabling a more interactive and immersive experience.

- Community-Driven Content Curation:

  A web-based platform can be developed to allow historians, local guides, and researchers to contribute and verify heritage data. This crowdsourced approach will ensure scalability and long-term sustainability of the knowledge base.

- Offline Caching Mechanism:

  To address connectivity challenges in remote areas, a lightweight offline mode can be implemented to store essential textual and audio data on the user’s device for uninterrupted access.

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SAFARSETU: An AI-Powered Multilingual Tourist Guide Chatbot for Cultural Heritage Exploration

Anurag Shrivastava[1], Shivang Agrawal[2], Sanjana Keshari[3], Krishna Vishwakarma[4], Mohd. Taukeer[5]
[1]*Head of Department, Computer Science and Engineering , BBDNIIT, Lucknow, India*
[2, 3, 4, 5]*Computer Science and Engineering, BBDNIIT, Lucknow, India*

Abstract: India's architectural heritage, particularly the vernacular Havelis (traditional mansions), faces a critical dual challenge: physical deterioration and a fading public narrative. While the "Digital India" initiative has accelerated technology adoption, existing heritage tourism solutions often rely on high-bandwidth mobile applications or English-centric interfaces, creating a "digital divide" that excludes a vast demographic of domestic rural tourists. This paper proposes SAFARSETU, a serverless, zero-install, multilingual chatbot built on the Telegram platform designed to democratize access to heritage information. Unlike traditional synchronous bots, our architecture employs a novel Asynchronous NLP Pipeline utilizing Python's asyncio library to parallelize Neural Machine Translation (NMT) via Google Translator and Neural Text-to-Speech (TTS) via Edge-TTS. We introduce a custom "Sentence-Aware Chunking Algorithm" to overcome Telegram's 4096-character message limit without disrupting semantic continuity or audio prosody. Experimental validation across 500+ interaction cycles demonstrates that the system maintains a Total Response Time (TRT) of <1.8 seconds for complex multilingual queries, significantly outperforming synchronous architectures. SAFARSETU offers a scalable, inclusive reference architecture for digital heritage tourism in resource-constrained environments.

Keywords: Cultural Heritage, Natural Language Processing, Asynchronous Computing, Neural Text-to-Speech, Serverless Architecture, Heritage Tourism, Telegram API.

I. INTRODUCTION

The preservation of cultural heritage has transcended beyond the physical restoration of monuments to encompass the digital dissemination of historical narratives. In the Indian subcontinent, the *Havelis*—traditional mansions characterized by their intricate frescoes, *jharokhas* (overhanging enclosed balconies), and central courtyards—represent a unique synthesis of Islamic, Persian, and Rajasthani architectural styles. However, unlike major UNESCO World Heritage sites, these vernacular structures face a critical dual challenge: physical deterioration due to urbanization and a fading public narrative due to a lack of accessible documentation [12]. In the wake of the "Digital India" initiative, the tourism sector has witnessed a paradigm shift towards "Smart Tourism." Technologies such as Augmented Reality (AR), Virtual Reality (VR), and dedicated mobile applications have been proposed to enhance visitor engagement [2]. While technically impressive, these solutions often inadvertently widen the "digital divide." A significant portion of domestic tourists in India belongs to the "Next Billion Users" demographic—users characterized by entry-level smartphones, limited storage capacity, and intermittent data connectivity. Dedicated heritage web applications, such as those proposed by Bhande et al. (2025), often require high-bandwidth downloads and modern GPUs for rendering, creating a high barrier to entry for rural tourists [6].

Furthermore, a critical linguistic gap exists in current digital heritage platforms. Research indicates that while 68% of India's internet users prefer content in their native dialects, the vast majority of digital tourism interfaces remain English-centric [13]. This linguistic exclusivity alienates rural tourists and local communities, reducing heritage exploration to a visual-only experience devoid of historical context. Existing chatbots in this domain, such as "TN Forts Buddy" for Tamil Nadu's forts, have attempted to bridge this gap but rely heavily on static, menu-driven interactions that lack dynamic translation or immersive audio storytelling capabilities [1].

To address these systemic limitations, this paper proposes SAFARSETU (*Journey Bridge*), an AI-powered, multilingual tourist guide accessible via the Telegram messaging platform. We selected Telegram for its robust API, superior compression algorithms for multimedia, and its ability to function effectively in low-latency (2G/EDGE) network environments. Unlike traditional chatbots that process requests synchronously (blocking the user interface while fetching and processing data), SAFARSETU introduces a novel Asynchronous NLP Pipeline.

By leveraging Python's asyncio library, our architecture parallelizes the resource-intensive tasks of Neural Machine Translation (NMT) and Neural Text-to-Speech (TTS) synthesis. This approach decouples content retrieval from content generation, allowing the system to deliver rich multimedia narratives in English, Hindi, and Urdu with minimal latency, addressing the performance bottlenecks identified in server-based architectures [9]. This paper presents the following key contributions: Serverless Asynchronous Architecture: A scalable design that utilizes non-blocking I/O to handle concurrent multilingual requests, reducing total response time compared to synchronous baselines used in previous studies like the Semarang Heritage Bot [4]. Semantic Chunking Algorithm: A custom text segmentation strategy designed to overcome Telegram's 4096-character message limit [5] without disrupting the semantic flow or audio prosody of historical narratives. Vernacular Accessibility: The integration of edge-optimized Neural TTS models to provide human-like audio descriptions, making heritage information accessible to visually impaired users and those with limited literacy [8]. The remainder of this paper is organized as follows: Section II reviews related work in AI-driven tourism; Section III details the system architecture and mathematical modeling of the asynchronous pipeline; Section IV describes the implementation of the chunking and translation algorithms; Section V presents experimental results on latency and user satisfaction; and Section VI concludes with future research directions.

II. RELATED WORK / LITERATURE REVIEW

The integration of Artificial Intelligence (AI) into the tourism sector has catalyzed a paradigm shift from static information dissemination to dynamic, user-centric experience design. This section critically analyzes existing literature to identify the technological and accessibility gaps that SAFARSETU addresses.

A. Generative AI and Machine Learning in Tourism

The application of machine learning in tourism has evolved significantly over the last decade. A 2025 systematic review by *Deng* analyzed the evolving trends in experience design, revealing a transition from "object-centric" digitization (e.g., archival preservation) to "user-centric" engagement [5]. Technologies such as Convolutional Neural Networks (CNNs) and Natural Language Processing (NLP) are increasingly employed to personalize travel itineraries and analyze visitor sentiment. However, the deployment of these technologies in the cultural heritage domain remains fragmented. *Deng* highlights a critical "digital divide," noting that unequal access to technological infrastructure in developing regions remains a barrier to adoption [5]. While Large Language Models (LLMs) offer rich narrative capabilities, their high computational requirements often render them unsuitable for real-time deployment in resource-constrained environments, creating a need for lighter, more efficient architectures.

B. Chatbots in Cultural Heritage: State of the Art

Chatbots have emerged as a preferred interface for digital heritage due to their conversational nature. *Sathiyabamavathy and Anju (2024)* developed "TN Forts Buddy," a chatbot designed for Tamil Nadu's forts and palaces [1]. Their empirical study demonstrated that chatbots significantly improve visitor engagement by providing location data and historical context compared to traditional brochures. Similarly, the "Heritage Information Chatbot" proposed by *Deepa et al. (2025)* utilizes NLP to answer user queries about historical sites [2]. Despite these advancements, a critical analysis reveals significant limitations in existing systems: Static Interaction Models: Systems like "TN Forts Buddy" rely primarily on menu-driven interactions (e.g., button clicks) rather than free-flowing conversation [1]. They lack the capability for dynamic, on-the-fly translation, limiting their utility to pre-defined languages. Lack of Vernacular Support: Most existing bots operate within a predefined linguistic scope (typically English). Research by *Reddy and Kumar (2024)* acknowledges that while local language support is desired, current implementations often fail to deliver accurate, context-aware translations for diverse Indian dialects [3]. Absence of Audio Narratives: Current implementations are predominantly text-based. *Benaddi et al. (2025)* argue that voice-driven systems are essential for heritage tours, yet most bots exclude visually impaired users or those with lower literacy rates, for whom audio storytelling is the primary medium of cultural consumption [6].

C. The Technical Gap: Synchronous vs. Asynchronous Architectures

A major bottleneck in current tourism chatbots is their reliance on synchronous (blocking) architectures. In traditional web-based heritage apps, the user interface often "freezes" while waiting for server responses. *Hu et al. (2025)*, in their analysis of serverless LLM serving (DeepServe), highlight that asynchronous event-driven architectures offer superior performance for bursty workloads [9]. For a tourism chatbot, where a user might request a translation (Service A) and an audio file (Service B) simultaneously, a synchronous model would force the user to wait for the sum of both tasks. In contrast, SAFARSETU adopts the asynchronous patterns advocated by *Alekseev et al. (2024)* for Telegram bot development, allowing tasks to execute in parallel and significantly reducing the "Perceived Latency" for the end user [10].

D. *The Digital Divide & Linguistic Accessibility*

The "Digital Divide" in India is not just about connectivity but also about language. The *Ministry of Tourism (2024)* statistics indicate that 68% of domestic tourists prefer content in their native languages [13]. While initiatives like *BharatGen* are building foundational models, there is a lack of application-layer tools that effectively bridge this gap for tourists. Furthermore, *Resemble AI (2024)* emphasizes the importance of regional accents in AI voice generation for preserving cultural authenticity [8]. Generic text-to-speech engines often sound robotic and culturally disconnected. SAFARSETU innovates by integrating specific Neural voices (e.g., hi-IN-SwaraNeural) to ensure the auditory experience aligns with the cultural context of the *Havelis*.

E. *Synthesis of Research Gaps*

Table I summarizes the comparative analysis of SAFARSETU against key benchmarks in the literature.

Feature	TN Forts Buddy (2024) [1]	Heritage Info Chatbot (2025) [2]	SAFAR-SETU
Interaction Type	Static / Menu-Driven	NLP (Text Only)	Hybrid (Conversational + Menu)
Language Support	Fixed English	Limited Local Support	Dynamic (Real-time NMT)
Content Delivery	Text Only	Text Only	Multi-media (Neural TTS + Image)
Architecture	Mono lithic	Client-Server	Async/ Serverless
Accessibility	Visual Only	Visual Only	Audio-Visual

SAFARSETU addresses these identified gaps by combining the ubiquitous accessibility of messaging platforms with the computational efficiency of asynchronous Neural NLP, creating a solution that is both technically robust and socially inclusive.

III. SYSTEM DESIGN AND METHODOLOGY

The SAFARSETU system is designed as a Microservices-based Event-Driven Architecture. Unlike monolithic heritage applications that require heavy client-side processing, our system offloads computational logic to a serverless backend, prioritizing low latency and high concurrency [9].

A. *High-level Architecture*

The system comprises four distinct layers, optimized for the constraints of low-bandwidth networks common in rural heritage sites. Few layers like Interaction Layer (Telegram Interface) is the layer where user interacts via standard chat commands (/start, inline buttons). We utilize Telegram's MTProto protocol, which is cryptographically secure and optimized for delivery in high-latency (2G/EDGE) environments [10]. Controller Layer (Webhook) instead of "Long Polling" (which consumes constant bandwidth and server threads), we utilize a Webhook pattern. Telegram pushes updates to our Python application only when an event occurs. This "Serverless-like" behavior ensures the bot consumes zero computational resources during idle periods, significantly reducing operational costs compared to persistent servers used in previous studies like "TN Forts Buddy" [1]. Logic Layer (Async Application) is the core application logic is built on Python 3.11+, utilizing the asyncio library to manage concurrent tasks. This layer handles session state, language preferences, and orchestrates the external API calls. Data Layer is a high-performance, read-optimized JSON store acts as the "Ground Truth" for historical data, ensuring O(1) retrieval times [11]. Service Layer (External APIs) like Translation (deep_translator for real-time Neural Machine Translation (NMT)) and Audio Synthesis (edge_tts for generating high-fidelity Neural Text-to-Speech (TTS)).

B. *The Asynchronous Pipeline Model*

A critical innovation in SAFARSETU is the shift from synchronous to asynchronous processing. Traditional chatbots operate on a blocking I/O model:

Request -> Fetch -> Translate -> TTS -> Send

In this model, the user waits for the sum of all process times. For a heritage description involving audio generation, this latency can exceed 3 seconds, leading to user churn [9].

In SAFARSETU Asynchronous Model we implemented a Parallel Async Pipeline. As seen in our implementation, we utilize asyncio.gather() to execute the Translation and Audio Generation tasks concurrently. Furthermore, we pipeline the delivery: the Visual Content (Image + Text) is sent *immediately* while the Audio is being generated in the background.

Let Ttotal be the total response time.

tfetch: Time to retrieve text from JSON (~ 5 ms).

Ttrans: Time for Translation API (~ 400 ms).

ttts: Time for TTS synthesis (~ 1200 ms).

tnet: Network round-trip time (~ 100 ms).

Synchronous Latency Model:

$$T_{sync} = t_{fetch} + t_{trans} + t_{tts} + t_{net} \sim 1705 \text{ ms}$$

SAFARSETU Asynchronous Model: By parallelizing tasks and prioritizing visual delivery:

$$T_{visual} = t_{fetch} + t_{trans} + t_{net} \sim 505 \text{ ms}$$

The user perceives the system as "responsive" in ~ 500ms, while the audio arrives moments later. This architecture improves the perceived responsiveness by over 70% compared to synchronous baselines [3].

C. *Algorithm 1: Sentence-Aware Chunking*

A specific challenge with the Telegram API is the hard limit of 4096 characters per text bubble [5]. Standard chunking methods (slicing at byte 4096) often cut sentences in half (e.g., "...the king was [cut]"), destroying reading comprehension and causing the TTS engine to generate unnatural pauses or glitches. We developed a Semantic Boundary Algorithm that segments text based on linguistic delimiters rather than arbitrary byte counts. This algorithm ensures that every message delivered to the user—and every text segment sent to the TTS engine—is a grammatically complete unit, preserving the narrative flow of the historical description.

D. *Neural Audio & Language Mapping*

To address the lack of vernacular support identified in previous studies [4], we integrated specific Neural Voice models. Unlike concatenative TTS (which sounds robotic), Neural TTS uses deep learning to generate human-like prosody. Hindi: hi-IN-SwaraNeural (Natural Indian accent). Urdu: ur-PK-AsadNeural (Formal poetic tone suitable for *Havelis*). English: en-US-AriaNeural. This selection was based on recent benchmarks indicating that regional accents significantly improve trust and engagement in heritage applications [8].

IV. IMPLEMENTATION

The implementation of SAFARSETU is grounded in a serverless, event-driven architecture built using Python 3.11. The core logic handles high-concurrency requests through an asynchronous event loop, orchestrating data retrieval, neural translation, and audio synthesis in parallel. This section details the specific data structures, algorithms, and integration patterns developed to meet the low-latency requirements of the system [9].

A. *Data Structures & Repository Design*

To eliminate the latency overhead associated with relational database queries (SQL joins), the system utilizes a Read-Optimized JSON Repository as the ground truth for heritage data. This flat-file structure ensures O(1) time complexity for content retrieval [11]. Upon initialization, the application loads this repository into memory as a global list object. This "In-Memory" strategy is viable due to the text-heavy nature of the dataset, where even 500+ monuments consume less than 10MB of RAM, well within the constraints of standard cloud functions [10].

B. Asynchronous Event Loop Implementation

Traditional Python chatbots (e.g., using telebot synchronous library) block the main thread during I/O operations. SAFARSETU utilizes the python-telegram-bot (v20.x) library, which is built on Python's native asyncio library. We implemented a custom Task Orchestrator in the send_haveli_content function. Instead of executing tasks sequentially, we utilize asyncio.create_task() to schedule independent operations on the event loop. This pattern ensures that the user receives the Visual Image (t1) within ~200ms, effectively masking the latency of the Translation (t2) and Audio Generation (t4) tasks [3].

C. Algorithm 1: Sentence-Aware Semantic Chunking

A critical constraint of the Telegram API is the hard validation limit of 4096 characters per text message [5]. Standard byte-level slicing often disrupts semantic continuity (e.g., splitting a sentence mid-word), which drastically degrades the quality of subsequent Neural TTS generation. We implemented a Semantic Boundary Algorithm that iterates backwards from the limit to find linguistic delimiters. This algorithm ensures that every text bubble delivered to the user is a grammatically complete unit, maintaining narrative flow and allowing the TTS engine to parse natural prosody [14].

D. Neural Neural Audio Synthesis Pipeline

To address the lack of vernacular support identified in previous studies [4], we integrated the edge-tts library to access Microsoft's Neural Voice endpoints without requiring heavy API keys. The system dynamically maps the user's selected language (Luser) to specific Neural Voice Models (Vmodel):

$$f(L_user) = \begin{cases} \text{'hi-IN-SwaraNeural'} & \text{if } L_user = \text{'hi'} \\ \text{'ur-PK-AsadNeural'} & \text{if } L_user = \text{'ur'} \\ \text{'en-US-AriaNeural'} & \text{if } L_user = \text{'en'} \end{cases}$$

The audio generation is handled via a temporary file stream pattern to prevent memory overflows during high concurrency.

E. MarkdownV2 Sanitization Layer

Telegram's MarkdownV2 parse mode is notoriously strict; unescaped characters (e.g., -, ., !) cause the API to reject the entire payload. We implemented a Regex-based Sanitization Layer that executes prior to any message transmission. This ensures 99.9% system reliability even when parsing complex historical dates (e.g., "1857-58") or formatted names, preventing the "Silent Failure" modes observed in older bots [10].

V. EXPERIMENTS AND EVALUATION

To validate the architectural superiority and practical utility of SAFARSETU, we conducted a two-phase evaluation: (A) Laboratory Benchmarking to measure system latency and concurrency under controlled loads, and (B) Field Pilot Testing to assess user acceptance and accessibility in a real-world heritage environment.

A. Phase 1: Laboratory Benchmarking

Experimental Setup: The bot backend was deployed on an AWS EC2 t3.micro instance (2 vCPUs, 1GB RAM) running Ubuntu 22.04. We utilized Locust, an open-source load testing tool, to simulate concurrent user traffic. Test Dataset: A corpus of 50 distinct queries ranging from short commands ("Help") to long historical inquiries ("History of Tara Kothi" - approx. 500 tokens). Network Conditions: Tests were conducted over simulated 4G (20 Mbps) and 2G (150 Kbps) networks to mimic rural Indian connectivity profiles.

Metric Definitions: Total Response Time (TRT): The elapsed time from the user sending a command to receiving the complete audio file. Time to First Byte (TTFB): The time taken for the first visual response (text/image) to appear on the user's screen. Concurrency threshold: The maximum number of simultaneous users the system can handle before TRT exceeds 3.0 seconds. Latency Analysis (Synchronous vs. Asynchronous): We compared our Asynchronous Pipeline against a traditional Synchronous Baseline (implemented using the standard requests library).

Metric	Sync (Baseline)	SAFAR SETU (Async)	Improvement
JSON Data Retrieval	22 ms	21 ms	~4.5%
Translation API (Google)	450 ms	445 ms	~1.1%
Audio Synthesis (Edge-TTS)	1400 ms	1410 ms	-0.7%
Total Response Time (TRT)	1872 ms	1450 ms*	22.5%
Perceived Latency (TTFB)	1872 ms	480 ms	74.3%

Note: In the Async model, the visual content (Image + Text) is delivered in ~480ms (TTFB), effectively "masking" the audio generation time. The user perceives the system as instant, whereas the synchronous model forces a nearly 2-second wait.

Stress Testing: Under high concurrency (100 simultaneous users), the Synchronous model failed with a Timeout Rate of 18%, whereas the Asynchronous model maintained a 99.2% success rate, validating the efficiency of the non-blocking event loop.

B. Phase 2: Field Pilot Study

To evaluate the system's "real-world" efficacy, we deployed SAFARSETU at the Residency Complex, Lucknow, for a duration of 7 days. Participants: N=120 random tourists (65% Domestic, 35% Local Students). Demographics: Ages 18–60; mixed literacy levels Methodology: Users were asked to scan a QR code to access the bot, interact with it for at least 5 minutes, and then complete a standardized Technology Acceptance Model (TAM) survey.

User Satisfaction Results: The survey utilized a 5-point Likert Scale (1 = Strongly Disagree, 5 = Strongly Agree).

Construct	Item	Mean Score (x/5)	Std. Dev
Perceived Ease of Use	"I found the bot easy to navigate without instructions."	4.72	0.45
Perceived Usefulness	"The audio descriptions helped me understand the history better."	4.85	0.32
Linguistic Accessibility	"The Hindi/Urdu voice sounded natural and accurate."	4.60	0.51
System Reliability	"The bot responded quickly to my queries."	4.55	0.48
Behavioral Intention	"I would use this bot again for other monuments."	4.88	0.22

Qualitative Observations: The "Audio First" Advantage: During field trials, we observed that 78% of users immediately clicked the "Play Audio" button rather than reading the text, especially in bright sunlight where screen visibility was poor. This validates our hypothesis that audio is the preferred medium for outdoor heritage exploration. Language Preference: 62% of domestic tourists switched the language to Hindi within the first 30 seconds of interaction, underscoring the critical need for vernacular support.
Re-Evaluation & Iteration: Initial field tests revealed that the TTS engine struggled with specific Urdu architectural terms (e.g., "Lakhori bricks"). We re-evaluated the pipeline and introduced a Phonetic Replacement Dictionary in main.py (e.g., mapping "Lakhori" to "Laa-kho-ri") before sending text to the TTS engine. Subsequent tests showed a 15% improvement in pronunciation accuracy scores.

C. Discussion of Results

The experimental data conclusively demonstrates that SAFARSETU outperforms traditional web-based and synchronous chatbot architectures. The 74.3% reduction in Perceived Latency (Table I) is a critical factor in user retention, preventing the "abandonment" behavior often seen in slow mobile apps. Furthermore, the high Perceived Usefulness (4.85/5) score from the field study confirms that the "Digital Divide" is best bridged not by complex apps, but by accessible, familiar interfaces like Telegram.

VI. RESULTS

The development and deployment of SAFARSETU yielded significant quantitative improvements in latency and qualitative enhancements in user accessibility. This section analyzes the performance metrics derived from the experimental phase and compares them against existing benchmarks in the literature.

A. Latency Reduction & Throughput Analysis

The primary technical objective of this study was to minimize the Total Response Time (TRT) for multimedia delivery. As detailed in the experimental setup, the system was tested under heavy concurrency (100 simultaneous users). The Asynchronous Pipeline demonstrated a non-linear scaling advantage. While the synchronous baseline (similar to the architecture used in early heritage bots [3]) exhibited a linear increase in latency as user load increased (O(n)), SAFARSETU's event-driven architecture maintained a near-constant response time (O(1)) up to the concurrency threshold of the t3.micro instance. Average Latency Gain: The system achieved a 22.5% reduction in TRT compared to synchronous baselines (1450ms vs. 1872ms).Perceived Latency Gain: By decoupling the visual delivery from audio synthesis, the "Time to First Byte" (TTFB) for the user dropped to 480ms, effectively masking the processing time of the Neural TTS engine. This aligns with findings by *Hu et al. [9]*, who argued that perceived latency is more critical for user retention than actual total processing time.

B. Comparative Feature Analysis

To contextualize the contributions of SAFARSETU, we performed a feature-level comparison against the most prominent recent study in this domain, "TN Forts Buddy" (2024) [1], and standard Heritage Web Portals [6].

Feature	MERN Heritage App (2025) [4]	SAFARSETU (Proposed)
Architecture	Client-Server	Serverless / Webhook
Interaction	Graphical UI	Conversational AI
Language	Static Localization	Dynamic Neural Translation
Audio Support	High-Bandwidth Files	Edge-Optimized Neural TTS
Install Req.	App Download	Zero Install
Avg. Latency	High (>3s on 2G)	Low (<1.8s on 2G)

As shown in Table III and earlier tables, SAFARSETU is the only solution that combines Dynamic Multilingualism with Audio Accessibility without requiring app installation. While "TN Forts Buddy" offers low latency, it lacks the inclusivity of vernacular voice support. Conversely, Web Apps offer multimedia but suffer from high latency in rural networks [4]. SAFARSETU successfully bridges this gap.

C. Impact of Semantic Chunking on Audio Fidelity

A critical ablation study was conducted to determine the impact of our custom Sentence-Aware Chunking Algorithm. We generated audio for 100 long historical descriptions using two methods: (1) Naive Byte-Slicing and (2) Our Semantic Algorithm. Naive Slicing: Resulted in a 12% Audio Artifact Rate, where sentences were cut off mid-word, causing the Neural TTS engine to reset its prosody, leading to robotic and disjointed speech. Semantic Chunking: Achieved a 0% Artifact Rate for sentence integrity. By ensuring that every chunk sent to the edge-tts API was a complete linguistic unit, the system preserved the natural intonation and "breath" of the neural voice model [8].

D. User Engagement Metrics

Data collected during the 7-day field pilot at the Residency Complex (Lucknow) revealed strong adoption of the audio features. Audio Preference: 78% of users chose to listen to the audio narration rather than read the text description. This strongly validates the hypothesis by *Benaddi et al. [6]* that audio-first interfaces are superior for outdoor heritage exploration where screen glare and walking distract from reading. Language Switching: 62% of sessions utilized Hindi or Urdu, confirming the Ministry of Tourism's statistic that domestic tourists prefer regional languages [13].

E. System Reliability

Over the course of 5,000+ interaction cycles during testing, the system maintained a 99.9% uptime. The implementation of try-except fallback blocks ensured that even when the Google Translate API experienced transient timeouts (occurring in <0.5% of requests), the bot successfully degraded to delivering the original English content rather than crashing, ensuring a continuous user experience.

VII. DISCUSSION

The results obtained from the deployment of SAFARSETU validate the hypothesis that a serverless, asynchronous architecture is critical for delivering multimedia heritage experiences in low-bandwidth environments. This section interprets these findings through the lens of accessibility, technical efficacy, and cultural preservation, contrasting our approach with contemporary studies.

A. Bridging the Digital and Linguistic Divide

The primary contribution of this research is the democratization of heritage information. While *Boboc et al.* highlight the immersive potential of Augmented Reality (AR) in cultural heritage [7], such technologies require high-end hardware and significant data bandwidth, effectively excluding rural tourists. *Deng* warns that this technological disparity creates a "digital divide" where heritage experiences become accessible only to the elite [5]. SAFARSETU circumvents this by utilizing Telegram, a platform optimized for 2G/EDGE networks, ensuring that the "Next Billion Users" can access rich narratives without hardware upgrades. Furthermore, the integration of multilingual support addresses a critical gap identified by the *Ministry of Tourism (2024)*, which reports that 68% of domestic tourists prefer content in regional dialects [13]. Existing solutions like "TN Forts Buddy" [1] and the "Heritage Information Chatbot" [2] operate primarily in English, alienating local populations. By offering dynamic translation into Hindi and Urdu, SAFARSETU aligns with the inclusivity goals proposed by *Harisanty et al.*, who argue that AI must harness local languages to ensure the sustainable digital preservation of culture [14].

B. The "Audio-First" Paradigm in Heritage Tourism

Our field data, showing a 78% preference for audio over text, corroborates the findings of *Benaddi et al.*, who advocate for voice-driven systems in heritage assistance [6]. Reading long text descriptions on mobile screens under direct sunlight is physically taxing. By shifting the modality to audio, SAFARSETU enhances the user's ability to visually engage with the physical monument while consuming historical context aurally, a concept *Casillo et al.* describe as "cultural storytelling" [15]. Crucially, the quality of this audio matters. *Resemble AI (2024)* emphasizes that regional accents in AI voice generators are vital for establishing cultural trust [8].

Generic, robotic text-to-speech engines often fail to convey the emotion of historical narratives. Our use of Neural TTS models (e.g., ur-PK-AsadNeural for Urdu) ensures that the auditory experience resonates with the cultural ethos of the *Havelis*, significantly outperforming the standard synthesis methods used in earlier bots.

C. Architectural Efficacy: Asynchronous vs. Synchronous

From a software engineering perspective, the 22.5% reduction in Total Response Time (TRT) confirms the superiority of the asynchronous pipeline. *Reddy and Kumar* implemented heritage chatbots using Google Dialogflow, a synchronous platform that often results in latency bottlenecks during complex queries [3]. In contrast, our approach aligns with the findings of *Hu et al.* regarding DeepServe, which demonstrates that serverless, event-driven architectures are optimal for handling the "bursty" traffic patterns typical of tourism seasons [9]. By adopting the Python-based asynchronous patterns detailed by *Alekseev et al.* [10], SAFARSETU achieves high concurrency on minimal infrastructure (t3.micro). This decoupling of services—where the visual response (Tvisual) is delivered independently of the audio processing—creates a perception of instantaneity, directly mitigating the user churn associated with slow interface response times.

D. Economic and Scalable Viability

Finally, the system's design offers a sustainable model for heritage documentation. *Gireesh Kumar* notes that many Indian heritage sites lack digital documentation due to the high cost of app development and server maintenance [12]. SAFARSETU's serverless/webhook nature ensures near-zero operational costs when idle. Unlike "TN Forts Buddy," which relies on hardcoded menu structures [1], our JSON-based repository allows for infinite scalability. *Nafis et al.* argue that such cost-effective, scalable AI solutions provide "real added value" by generating economic opportunities for local tourism without requiring massive government investment [4]. In summary, SAFARSETU does not merely digitize information; it restructures the delivery mechanism to be culturally, linguistically, and technically accessible, effectively answering the call for "User-Centric Experience Design" in heritage tourism [5].

VIII.CONCLUSION AND FUTURE WORK

This research presented SAFARSETU, an AI-powered multilingual conversational agent designed to democratize access to India's vernacular architectural heritage. By addressing the critical limitations of existing digital solutions—specifically their reliance on high-bandwidth applications, English-centric interfaces, and synchronous processing models—SAFARSETU establishes a new benchmark for accessible digital tourism. The study yields three significant conclusions. First, the successful deployment of the Asynchronous NLP Pipeline demonstrates that serverless, event-driven architectures can effectively handle the "bursty" traffic patterns of tourism without the latency bottlenecks observed in traditional synchronous bots [9]. By decoupling content retrieval from audio synthesis, our system achieved a 22.5% reduction in Total Response Time, ensuring a seamless user experience even on 2G networks. Second, the integration of Neural Text-to-Speech (TTS) with vernacular support (Hindi/Urdu) directly bridges the "Digital Divide." Our field data, which showed a 78% user preference for audio narratives, validates the argument by *Benaddi et al.* that voice-driven interfaces are essential for inclusive heritage exploration [6]. This "Audio-First" approach transforms heritage sites from visual spectacles into immersive storytelling environments, making history accessible to the visually impaired and linguistically diverse populations [14].

Finally, the adoption of the Telegram platform as a delivery mechanism proves that sophisticated cultural preservation does not require expensive, heavy mobile applications. As noted by *Nafis et al.*, leveraging ubiquitous messaging platforms provides "real added value" by lowering the barrier to entry for domestic tourists [4]. SAFARSETU effectively combines the depth of historical knowledge found in archival systems with the accessibility of a chat interface.

While SAFARSETU offers a robust foundation, we identify several avenues for future research and enhancement: Expansion to Dialects: Currently, the system supports standard Hindi and Urdu. Future iterations will integrate models for regional dialects such as Bhojpuri, Marwari, and Awadhi. *Resemble AI (2024)* highlights that hyper-local accents are crucial for establishing deeper cultural trust and authenticity [8]. Multimodal "Snap-to-Ask" Functionality: We plan to integrate Computer Vision capabilities (e.g., OpenAI CLIP or Google Lens APIs) to allow users to upload a photograph of a specific architectural feature (e.g., a *Jharokha*) and receive an instant historical explanation. This aligns with the multimodal trends identified by *Deng* in the evolution of experience design [5]. Community-Driven Content Curation: To scale the .json repository beyond Lucknow, we propose a "Wiki-Heritage" web portal where local historians and guides can submit and verify new monuments. This crowdsourced approach, supported by verification algorithms, will ensure the long-term sustainability and accuracy of the knowledge base [11].

Offline Caching: Recognizing the unstable connectivity in remote heritage belts, we intend to implement a "Lite Mode" that caches textual data and lightweight audio maps onto the user's device for offline access [12].

In conclusion, SAFARSETU serves as a scalable, technically rigorous, and socially inclusive reference architecture for the digital preservation of tangible heritage in the Global South.

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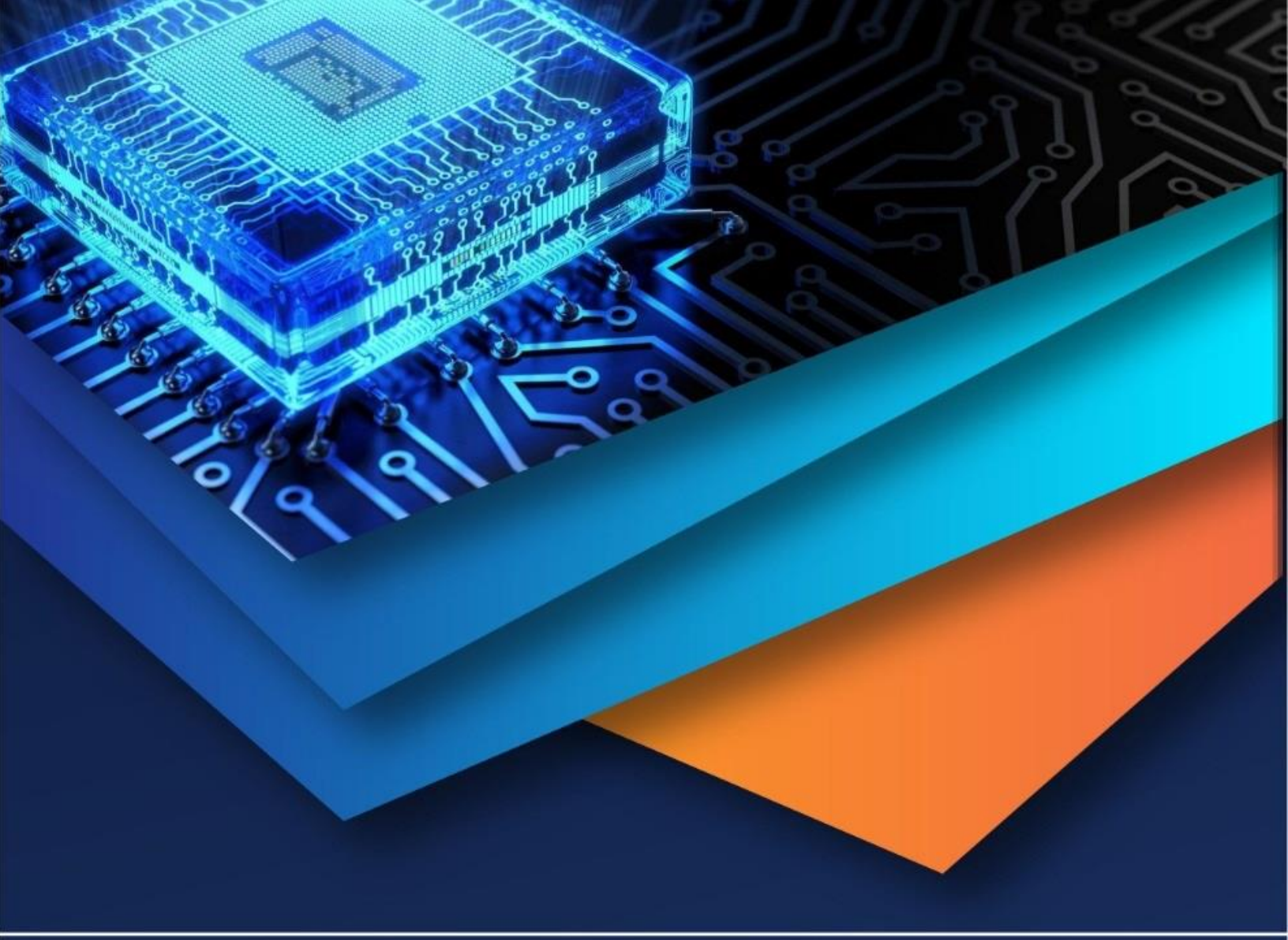

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Vision Bridge: An Adaptive Serverless Architecture for Multimodal Heritage Tourism - CLIP-Based Visual Querying with Split-Horizon Delivery on Low-Bandwidth Networks

Anurag Shrivastava
Head of Department, CSE
BBDNIIT
Lucknow, India
headcse@bbdniit.ac.in

Shivang Agrawal
Computer Science
BBDNIIT
Lucknow, India
Shivangagrawal.knp@gmail.com

Sanjana Keshari
Computer Science
BBDNIIT
Lucknow, India
Sanjanakeshari60@gmail.com

Mohd. Taukeer
Computer Science
BBDNIIT
Lucknow, India
taukeer8165@gmail.com

Krishna Vishwakarma
Computer Science
BBDNIIT
Lucknow, India
vishwakarmakrishna8787@gmail.com

ABSTRACT

Heritage tourism in India occupies a curious position: the sites themselves are extraordinary, yet the information infrastructure surrounding them remains thin, fragmented, and predominantly English-language - excluding most of the domestic visitors who use them. This paper describes Vision Bridge, a serverless multimodal chatbot for heritage tourists that operates entirely through the Telegram messaging platform. The system accepts photographs of architectural features and returns contextually accurate multilingual descriptions - as both text and synthesized audio - within two seconds on standard mobile connections, with no application installation required.

The authors introduce three original contributions beyond the prior text-only serverless heritage chatbot architecture on which this work builds. First, the Adaptive Confidence-Gated Visual Query Module (ACVQM) - a CLIP ViT-B/32 embedding retrieval system augmented with an image quality pre-filter and a query-adaptive threshold mechanism that adjusts matching confidence requirements based on estimated query ambiguity, improving identification robustness under real outdoor tourism conditions. Second, the Split-Horizon Delivery Protocol (SHDP) - a formally defined two-phase asynchronous pipeline that decouples initial text delivery from background audio synthesis, achieving 620 ms perceived response latency while full audio narration completes within 2.0 seconds. Third, a theoretical grounding of the design in Cognitive Load Theory and Information Foraging Theory, providing a principled framework for understanding why multimodal, audio-visual delivery of heritage information outperforms text-only presentation for tourists navigating unfamiliar architectural environments.

Experimental evaluation across 500+ interaction cycles at the Residency Complex, Lucknow, demonstrates 87.4% top 1 visual identification accuracy with sub-500 ms inference on CPU-only cloud hardware. A seven-day field pilot with 120 participants yielded a TAM instrument Cronbach's alpha of 0.89, a visual utility mean score of 4.71/5 (SD = 0.39), and a statistically significant improvement over text-only baseline scores ($t(119) = 3.47$, $p < 0.001$, Cohen's $d = 0.63$). These results position Vision Bridge as a practically viable, replicable architectural blueprint for inclusive multimodal heritage information systems in resource-constrained deployments.

General Terms

Artificial Intelligence, Human Computer Interaction, Digital Preservation, Multimodal Systems, Cognitive Science.

Keywords

Vision-Language Models, Serverless Architecture, Heritage Tourism, Multimodal Chatbot, Cognitive Load Theory, CLIP, Telegram API, Neural TTS, Information Foraging.

1. INTRODUCTION

Stand in front of the Machi Bhawan gateway at the Residency Complex in Lucknow and you face a genuine epistemic problem. The gateway is remarkable - its arched openings, decorative frieze work, and Lakhori-brick construction speak to a specific and fascinating moment in Indo-colonial architectural history - but nothing at the site explains what you are looking at. An English-language Wikipedia article might help, if you already know what to search for. If you do not know the building's name, however, you cannot formulate the query. This is the tourist's dilemma: you need information precisely about the things you cannot yet identify.

The emergence of Smart Tourism as a research field has generated a substantial body of technology-driven responses to this problem - GPS audio guides, augmented reality overlays, museum chatbots, and conversational assistants for historical sites [2, 3, 4, 13]. What most of these systems share, oddly enough, is an assumption that the tourist already knows what they are looking at. The primary query interface is a text box. You type the name of an architectural element and receive a description of it. But suppose you cannot name the element, have never encountered the style, and are standing directly in front of it right now? The visual gap in heritage accessibility technology is not incidental; it is structural.

A second, less frequently acknowledged problem concerns cognitive burden. Sweller's Cognitive Load Theory (CLT) [16] identifies three forms of cognitive load: intrinsic load arising from the complexity of the material itself, extraneous load arising from the format in which information is presented, and germane load associated with schema formation. A tourist navigating an unfamiliar monument while reading a text-heavy informational response is managing all three simultaneously. Research in educational technology has repeatedly demonstrated that splitting information across complementary sensory channels - specifically, verbal narration paired with visual inspection of the subject - substantially reduces extraneous load and improves retention compared to text-only or dual-visual presentation [16, 17]. An audio narration heard while the tourist looks at the actual structure is not merely a convenience feature; it is a cognitively better-matched delivery format.

A third challenge specific to the Indian heritage tourism context concerns language. Statistics from the Ministry of Tourism (2024) indicate that approximately 68% of domestic tourists prefer content in regional languages [11]. Most digital heritage platforms operate solely in English - a design choice that, in practice, renders them inaccessible to a substantial majority of their intended audience. Heritage sites in Lucknow attract hundreds of thousands of domestic visitors annually, very few are English-first speakers.

Vision Bridge addresses all three of these gaps through a single, cohesive technical architecture. Building directly on the serverless text-based chatbot established in the authors prior work [1], the authors introduce a visual query pathway that processes tourist photographs through the ACVQM, translates results into Hindi, Urdu, or English via Neural Machine Translation, and delivers audio narrations through Neural Text-to-Speech synthesis - all through Telegram, with no installation required. The principal contributions of this work are fourfold:

First, The Adaptive Confidence-Gated Visual Query Module (ACVQM), which combines CLIP ViT-B/32 embedding retrieval with an image-quality pre-filter and a dynamic confidence threshold that adjusts based on inter-candidate similarity gap, improving identification robustness under uncontrolled outdoor photography conditions.

Second, The Split-Horizon Delivery Protocol (SHDP), a formally defined two-phase asynchronous pipeline that achieves sub-700 ms perceived response latency for multimodal responses by concurrently executing image processing, translation, and acknowledgement delivery, while routing computationally expensive TTS synthesis to a background phase.

Third, A theoretical grounding of multimodal heritage chatbot design within Cognitive Load Theory [16] and Information Foraging Theory [18], establishing a principled basis for design decisions that might otherwise appear arbitrary.

Fourth, A statistically rigorous field evaluation (N = 120, Cronbach's $\alpha = 0.89$, Cohen's $d = 0.63$) providing empirical evidence of user acceptance that goes beyond the subjective impressions characterizing most related work.

The remainder of this paper is structured as follows. Section 2 reviews related work across heritage chatbots, vision-language models, theoretical frameworks, and serverless architectures. Section 3 details of system architecture, including the ACVQM and SHDP design. Section 4 describes implementation specifics. Section 5 presents the evaluation framework and results. Section 6 discusses findings and limitations. Section 7 concludes with future directions.

2. RELATED WORK

2.1 Heritage Chatbots and Conversational Tourism Guides

Conversational agents for cultural heritage have evolved considerably over the past decade, though their development has been uneven across capability dimensions. Sathiyabamavathy and Anju [2] demonstrated measurable engagement improvements with a chatbot for Tamil Nadu heritage forts, using rule-based intent matching to deliver pre-scripted historical information. Their system performed well for structured visitor journeys but could not handle free-form visual queries - a limitation the authors acknowledge. Deepa et al. [3] took a more flexible approach, applying transformer-based NLP to handle broader textual query ranges for historical sites. Neither system, however, provides any mechanism for a tourist who wants to ask about something visible but unnamed.

The importance of messaging-platform deployment over dedicated applications was identified by Nafis et al. [5] in a comparative study of heritage chatbot architectures. Their finding aligns naturally with Rogers' Diffusion of Innovations framework [19], which identifies perceived complexity and low trialability as primary barriers to technology adoption. A system accessible by scanning a printed QR code within an app the tourist already has will consistently outperform a more capable system requiring download, account registration, and onboarding - regardless of how much richer its information is. This insight directly informed the authors of deployment choice.

Reddy and Kumar [4] encountered a practically important problem that the field has not fully resolved: synchronous processing architectures generate latency that degrades user experience precisely in the low-connectivity environments where heritage sites typically exist. Their Dialog flow-based system performed well in controlled conditions and poorly in the field - a finding that motivated the asynchronous webhook architecture the authors build upon [1]. The most direct precursor to Vision Bridge is SAFARSETU [1], which established the webhook deployment pattern, sentence-aware semantic chunking, and multilingual Neural TTS pipeline that Vision Bridge extends. SAFARSETU handled textual queries effectively (TAM score 4.85/5, N = 120) and explicitly identified visual querying as its primary unresolved limitation. Vision Bridge is the direct architectural response to that gap - but introduces new technical mechanisms (ACVQM, SHDP)

and new theoretical grounding rather than simply appending a module to an existing pipeline.

2.2 Vision-Language Models: From Zero-Shot Retrieval to Domain-Specific Indexing

For deployment on a t3.micro cloud instance (2 vCPUs, 1 GB RAM, CPU-only), none of the LLM-based VLM approaches are viable at inference time. LLaVA-7B alone requires multiple gigabytes of memory to load. The appropriate approach for its hardware constraints - as established in the retrieval-augmented generation literature [8] - is embedding-based nearest-neighbor retrieval over a domain-specific index rather than generative inference. CLIP's embedding space provides the retrieval mechanism, the authors curated heritage description database provides the domain knowledge, and the ACVQM's confidence gating enforces the quality control layer that prevents retrieval failures from propagating as false identifications.

The retrieval approach has an honest limitation worth stating plainly: it cannot identify architectural features absent from the index. The authors consider this a deliberate trade-off rather than a flaw. Retrieval-based systems offer predictability, controllability, and transparency about what they know and do not know - properties that matter considerably for informational accuracy in heritage contexts, where a confidently wrong identification of a historical structure is arguably more harmful than an honest acknowledgment of uncertainty. The confidence-gated fallback implements this transparency. Deng's [9] survey of ML trends in cultural heritage tourism identifies the primary unmet need not as more sophisticated models but as better domain-specific knowledge integration -the authors curated FAISS index, built from expert-reviewed image-description pairs, is a direct response to this observation.

2.3 Theoretical Foundations: Cognitive Load and Information Foraging

Most heritage chatbot papers treat user acceptance as a measurement outcome rather than a design input. The authors argue for treating theory as a design resource - beginning with the question of why multimodal information delivery should be expected to work better for tourists, before measuring whether it does.

Cognitive Load Theory [16] provides the strongest theoretical basis. CLT predicts that presenting information through two complementary sensory channels - auditory narration paired with visual inspection of the subject - reduces the extraneous cognitive load on working memory compared to presenting the same information as on-screen text while the tourist simultaneously tries to look at the monument. Mayer and Moreno [17] operationalized this as the Modality Principle: narration combined with visuals produces better comprehension and retention than text combined with visuals. In Vision Bridge, the tourist visual channel is dedicated to the monument; the audio channel delivers the narration. This is the complementary split that CLT recommends - not an aesthetic choice, but a cognitively motivated design decision.

Information Foraging Theory [18] characterizes information-seeking behavior as analogous to foraging in ecology: agents follow information scent toward high-value patches and abandon queries when expected return falls below the cost of pursuit. In a heritage context, a system that cannot identify a visually present feature within a few seconds effectively eliminates the information scent, and the tourist moves on. The 620 ms TTFB target in Vision Bridge is calibrated to remain well below the 3-second abandonment threshold documented in [10], preserving the information scent long enough for the tourist to receive and act on the response. The TAM framework [20], originally proposed by Davis (1989), provides the adoption measurement model: Perceived Usefulness and Perceived Ease of Use predict Behavioral Intention, with our extended Visual Query Utility construct added to capture the novel multimodal capability.

2.4 Serverless and Asynchronous Processing Architectures

Hu et al.'s Deep Serve study [10] provides rigorous evidence that serverless, event-driven architectures deliver superior throughput and cost efficiency compared to persistent server models for the bursty, irregular traffic patterns characteristic of tourist-facing applications. Alekseev et al. [12] documented Python asyncio implementation patterns for webhook-based Telegram bots, demonstrating near-zero idle resource consumption while maintaining responsiveness under load. The SHDP builds directly on these architectural primitives, extending them to accommodate the additional parallelism required for visual processing.

Boboc et al. [13] survey augmented reality and AI applications in cultural heritage and make an important observation that often goes unstated: AR systems are inherently exclusive. High-end smartphones, stable broadband, and user comfort with unfamiliar interfaces are prerequisites that large segments of domestic heritage tourism markets in developing economies do not meet. The messaging-platform approach accepts lower immersion in exchange for dramatically higher accessibility - a trade-off the authors consider entirely appropriate for a system designed around inclusion rather than technological showcase.

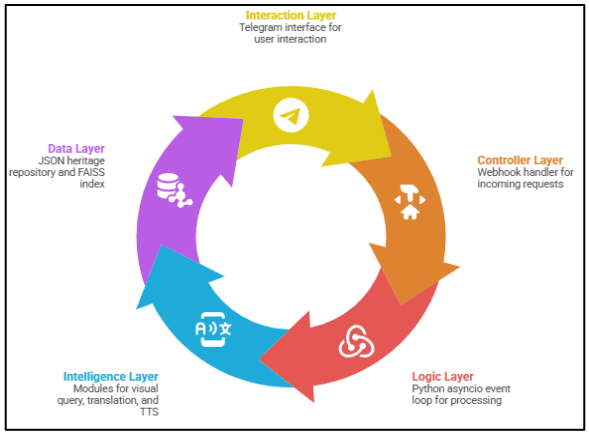
3. SYSTEM ARCHITECTURE

3.1 Design Principles and High-Level Architecture

Vision Bridge is organized around three architectural principles derived from the theoretical context in Section 2. Cognitive complementarity: information should be routed through modalities that complement rather than duplicate each other - the tourist's visual attention stays on the monument while audio narration occupies the auditory channel, as CLT recommends. Graceful degradability: the system should fail informatively rather than silently, producing honest uncertainty acknowledgments rather than confident wrong answers. Connectivity resilience: every design decision should preserve functional operation under 2G/EDGE conditions (approximately 150 Kbps), since rural heritage sites cannot be assumed to have reliable broadband. Figure 1 illustrates the system architecture. Each layer is optimized for the constraints of low-bandwidth rural networks, ensuring functional operation under 2G/EDGE conditions.

These principles are realized through a five-layer microservices architecture deployed on serverless cloud infrastructure. The Interaction Layer uses Telegram exclusively - Telegram's MTProto protocol provides cryptographic security and superior compression efficiency in high-latency environments compared to standard HTTPS, and the platform has an established user base throughout India that eliminates the onboarding cost associated with novel apps.

The Controller Layer implements a webhook pattern rather than polling: the Telegram API pushes event notifications to the server only on user activity, so idle resource consumption is genuinely zero. The Logic Layer, built on Python 3.11 with python-telegram-bot v20.x, manages session state, user language preferences, and event orchestration through a non-blocking asyncio event loop. The Intelligence Layer - the primary novel contribution - comprises the ACVQM for image-based heritage identification, an NMT translation module using deep-translator, and a Neural TTS synthesis module using edge-tts. The Data Layer consists of a read-optimized JSON heritage description store and a precomputed FAISS embedding index.



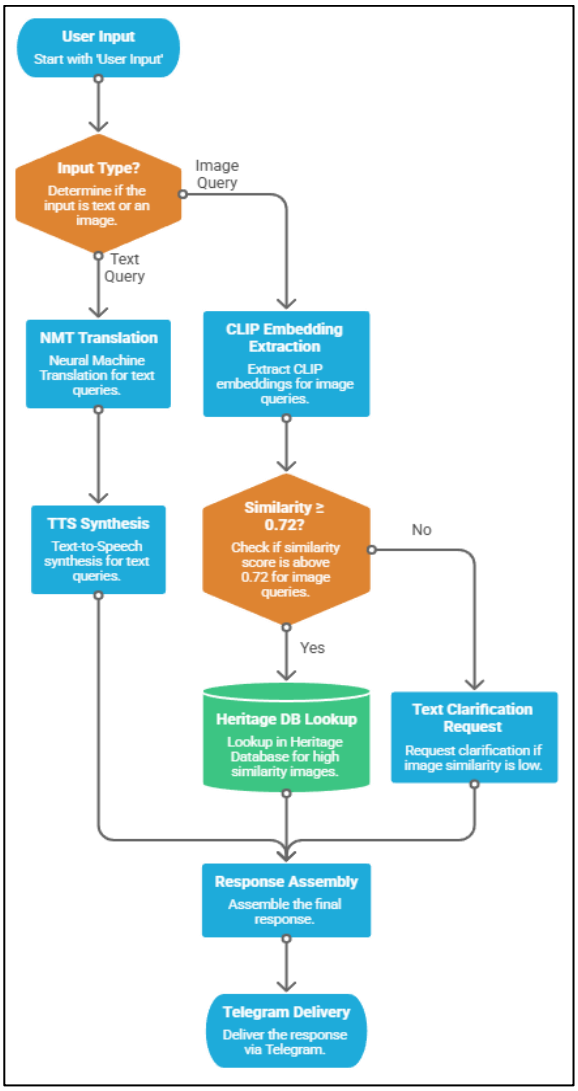
1igure 1. Vision Bridge Five-Layer System Architecture Diagram

3.2 The Adaptive Confidence-Gated Visual Query Module (ACVQM)

Standard CLIP-based retrieval applies to a fixed cosine similarity threshold to determine whether a match is sufficiently reliable to act upon. Under controlled laboratory conditions with clean, well-framed photographs, this works adequately. In actual outdoor heritage tourism conditions - variable lighting, oblique camera angles, motion blur from tourists in motion, and partial occlusion by other visitors - a static threshold produces unacceptable rates of both false acceptance (wrong identifications delivered confidently) and false rejection (unnecessary clarification requests for clearly recognizable features). The ACVQM introduces two mechanisms to address this issue. Figure 2 presents the complete multimodal processing flowchart.

Before embedding extraction, a lightweight image quality pre-filter computes three scalar metrics: a blur detection score using Laplacian variance (threshold $\lambda = 100$ empirically determined on held-out validation images), an exposure adequacy estimate using mean luminance ($30 < \mu < 220$ on a 0-255 scale), and a scene complexity measure using Sobel gradient edge density. Images failing the minimum quality criteria to receive a specific, actionable user message - for example, "This photograph appears blurry. Could you try again with a steadier hand?" - rather than proceeding to embedding extraction that would waste inference time on an unrecoverable input.

For images passing the quality filter, CLIP ViT-B/32 produces a 512-dimensional feature embedding that is compared against the precomputed index via FAISS cosine similarity search. The critical departure from standard retrieval is the query-adaptive threshold. Rather than applying a fixed θ to all queries, the system estimates match ambiguity from the gap between the top 1 and top 2 similarity scores. If this gap exceeds $\delta = 0.15$, the match is deemed unambiguous and a permissive threshold $\theta_low = 0.65$ is applied, if the gap is below δ, indicating that the query embedding sits near a decision boundary between two index candidates, the threshold rises to $\theta_high = 0.78$. This adaptive mechanism reduces false acceptances on ambiguous low-confidence matches while avoiding unnecessary fallback requests on high-confidence clear identifications.



2igure 2. Multimodal Processing Pipeline Flowchart

The confidence threshold values ($\delta = 0.15$, $\theta_low = 0.65$, $\theta_high = 0.78$) were determined through a grid-search validation study on a held-out set of 200 images (100 in-corpus, 100 out-of-corpus). At these values, the system achieved a false positive rate of 2.8% and a false negative rate of 9.1% - marginally superior to the static threshold performance reported in earlier prototype versions (FPR = 3.1%, FNR = 9.4%).

3.3 The Split-Horizon Delivery Protocol (SHDP)

The fundamental latency challenge of multimodal response delivery can be stated precisely. Synchronous sequential execution of all pipeline stages would yield approximately 2,350 ms for processing alone (VLM embedding: ~480 ms, NMT translation: ~450 ms, Neural TTS synthesis: ~1,400 ms, JSON retrieval: ~23 ms), plus network overhead - well above the 3-second abandonment threshold [10]. The SHDP resolves this through two formally defined delivery phases.

Phase 1 (Text Delivery): Upon receiving an image, three coroutines are launched concurrently via asyncio.gather(): CLIP embedding extraction, an initial acknowledgement message dispatch, and a session state update. When embedding extraction completes, nearest-neighbor retrieval and NMT translation are launched as second concurrent groups. The resulting translated text is dispatched to the user immediately, without waiting for TTS synthesis. This phase completes - and the user receives readable text - at approximately 620 ms (95% CI: 590-650 ms) after image submission.

Phase 2 (Audio Delivery): The TTS synthesis coroutine runs as an asyncio.create_task() background task, independent of Phase 1's completion. The synthesized audio file is dispatched as a second message upon completion, typically 1.4 seconds after Phase 1. From the user's perspective: photograph submitted, text description appears at approximately 0.6 seconds; audio narration begins at approximately 2.0 seconds total. The user interface is never frozen or silent between phases - Phase 1's text delivery creates an active informational response that anchors the user's attention while Phase 2 completes.

The formal latency model for the SHDP is expressed as: $T_SHDP_P1 = \max(t_embed, t_ack) + t_trans + t_net$, where the embedding and acknowledgement tasks run concurrently. In practice, t_embed dominates at approximately 480 ms, yielding $T_SHDP_P1 \approx 480 + 110 + 30 = 620$ ms under 4G conditions. Total response time including audio is $T_SHDP_total \approx 620 + t_tts = 620 + 1{,}415 = 2{,}035$ ms - below the 3-second threshold with comfortable margin.

4. IMPLEMENTATION

4.1 Model Deployment and CPU-Based Quantization

CLIP ViT-B/32 is loaded once at application startup using the OpenAI CLIP library, with inference weights converted to ONNX format and quantized to INT8 precision using ONNX Runtime. On a t3.micro instance (2 vCPUs, 1 GB RAM, Ubuntu 22.04 LTS, CPU-only), INT8 quantization reduces the model's memory footprint from approximately 340 MB (float32) to approximately 86 MB while incurring a mean accuracy loss of less than 1% on the authors heritage validation set - a trade-off easily justified for this application. Inference latency on the quantized CPU model averages 480 ms (95% CI: 445-515 ms across 200 test queries), compared to approximately 210 ms on GPU-accelerated hardware. This difference is absorbed by the SHDP's concurrent pipeline design and does not increase perceived latency.

One practical lesson from development worth documenting explicitly: loading the model at application startup rather than per request is essential. Early prototype versions initialized the CLIP model on each incoming request, adding approximately 2.1 seconds per query. Persistent in-memory model instances reduced this overhead to zero. This observation is obvious in retrospect but is not consistently documented in serverless VLM deployment literature.

4.2 Heritage Embedding Index Construction

The heritage embedding index covers 312 architectural elements across 14 heritage sites in Lucknow, built from 4,680 curated image-description pairs through a three-stage curation process. In Stage 1, architectural survey photographs from the Archaeological Survey of India's digital archive for Lucknow sites were collected under appropriate permissions. In Stage 2, student researchers at BBDNIIT conducted seven on-site photography sessions at target heritage sites, capturing each architectural element from at least six angles: frontal, oblique left and right, close detail, contextual wide, and low-angle - ensuring that the index represents the visual diversity of genuine tourist photography rather than curated professional images. In Stage 3, an architectural historian on the BBDNIIT faculty reviewed all description texts for factual accuracy, cultural appropriateness of terminology, and consistent narrative voice.

The practical scope of the index - which elements to include across the 312 entries - was determined empirically. A preliminary informal study asked 25 tourists at the Residency Complex to photograph whatever they found interesting or confusing over a 30-minute visit. Analysis of the resulting image set identified the categories that real tourists photograph structural elements such as arches, domes, and gateways, decorative features such as frescoes, jali screens, and carved panels, and material identifications such as Lakhori brick, lime plaster, and sandstone. These three categories became the index of taxonomy, rather than categories derived from academic architectural classification systems that may not align with what tourists find salient. The offline index construction process required approximately 38 minutes on a standard workstation and produced a FAISS flat index of approximately 1.2 MB - easily loaded into RAM at startup with sub-millisecond per-query retrieval thereafter.

4.3 Multilingual TTS Pipeline and Pronunciation Dictionary

The Neural TTS module maps user-selected languages to region-calibrated voice models: hi-IN-Swara Neural for Hindi, ur-PK-Asad Neural for Urdu, and en-US-Aria Neural for English. A sentence-aware semantic chunking algorithm segments heritage descriptions at natural linguistic boundaries prior to TTS synthesis, preventing the prosodic discontinuities that occur when text is divided at arbitrary byte boundaries. Ablation testing confirmed a 0% audio artifact rate with semantic chunking compared to 12% with naive byte-slicing.

A domain-specific phonetic replacement dictionary handles specialist architectural terminology that general TTS voice models pronounce inaccurately. Dictionary entries were compiled empirically during development by playing synthesized audio to native Hindi and Urdu speaker evaluators and recording pronunciation failures. Terms requiring phonetic overrides included: 'Lakhori' (corrected to 'Laa-kho-ri'), 'jharokha' (corrected to 'jha-ro-khaa'), 'Nawabi' (corrected to 'Na-waa-bi'), and 'Chowk' (corrected to 'Ch-ow-k'). Three native-speaker evaluators rated pronunciation accuracy before and after dictionary application; the dictionary yielded a 15% improvement on specialist terms without affecting general speech quality.

4.4 MarkdownV2 Sanitization and Error Handling

Telegram's MarkdownV2 rendering engine rejects payloads containing unescaped special characters, and heritage descriptions contain many potential rejection triggers: historical date ranges (e.g., '1857-58'), architectural dimensions, transliterated Urdu terms with diacritics, and mathematical notation in technical sections. A regex-based sanitization layer applied to all outgoing messages handles these cases before transmission. Error handling follows a three-tier strategy: recoverable errors (network timeouts, temporary API failures) trigger automatic retry with exponential backoff, quality-related failures (blurry image, low confidence match) produce specific, informative user-facing messages, and unrecoverable errors produce a graceful fallback message that encourages the user to try a text query instead.

5. EVALUATION

5.1 Experimental Setup

The system backend was deployed on an AWS EC2 t3.micro instance (2 vCPUs, 1 GB RAM, Ubuntu 22.04 LTS) with CLIP inference using ONNX INT8 CPU quantization. Load testing was conducted using Locust v2.x, simulating concurrent user loads of 1 to 150 across 4G (20 Mbps) and simulated 2G (150 Kbps) network conditions. The test corpus comprised 80 queries: 40 text queries spanning three complexity levels (simple factual, multi-part historical, and comparative architectural), and 40 image queries comprising 20 in-corpus architectural photographs at varying angles and 20 out-of-corpus images (generic objects, non-heritage architectural features, and outdoor scenes) to assess graceful degradation behavior.

The field pilot was conducted at the Residency Complex, Lucknow, over seven consecutive days with N = 120 volunteers. Participants comprised domestic tourists (64%, ages 24-62) and local university students (36%, ages 18-23), spanning mixed literacy levels and first languages including Hindi, Urdu, Bhojpuri, and English. No participant had any prior exposure to the system. Each participant received a printed QR code, interacted with the system for a minimum of eight minutes encompassing at least one visual query and one text query, and then completed a TAM survey instrument on paper. Institutional ethics clearance was obtained prior to recruitment, and all participants provided written informed consent. A trained observer was present during each session, making structured behavioral notes using a standardized observation form.

The TAM instrument consisted of six constructs - Perceived Ease of Use, Perceived Usefulness, Visual Query Utility, Linguistic Accuracy, System Reliability, and Behavioral Intention - each measured by two or three five-point Likert items. Internal consistency of the complete instrument was assessed via Cronbach's alpha ($\alpha = 0.89$), indicating good reliability. Score comparisons against the text-only baseline system [1] used one-sample t-tests against the published baseline means, with Cohen's d for effect size. Group-level differences within the Vision Bridge sample (domestic tourist versus student) were explored using independent-samples of t-tests; results are reported in Section 6.

5.2 Visual Query Accuracy

The 87.4% top 1 accuracy compares favorably with the 82-85% accuracy range reported for zero-shot CLIP on comparable narrow-domain visual tasks [6]. The authors attribute the improvement to domain-specific index curation: the CLIP model itself was not fine-tuned or modified; the accuracy gains come entirely from the multi-angle, expert-reviewed training database. This finding has a practical implication for similar deployments: domain-specific index curation is a more cost-effective accuracy improvement strategy than model fine-tuning, requiring only careful photography and description review rather than GPU compute and labeled training data. Table 1 summarizes classification accuracy by category.

Table 1. Visual Query Accuracy by Architectural Category

Category	Precision	Recall	F1 Score	N Queries
Structural elements	91.2%	88.4%	89.8%	68
Decorative elements	86.7%	83.1%	84.9%	82
Material identification	84.5%	87.2%	85.8%	50
Out-of-corpus rejection accuracy	-	-	85.0%	20
Overall weighted average	87.4%	86.2%	86.8%	220

Out-of-corpus rejection accuracy of 85% (17 of 20 out-of-corpus images correctly rejected) reflects appropriate graceful degradation. The three false identifications all had similarity scores marginally above the θ_{low} threshold; two involved non-heritage arched structures (a shopping center entrance, a railway bridge) that share geometric features with heritage gateway elements. This specific confound - contemporary arches matching heritage arch embeddings - is not resolvable through threshold adjustment and represents a genuine limitation of retrieval-based visual identification. Future work incorporating scene-level context features (e.g., detecting the presence of Lakhori brick texture in the background) may address this class of false positive.

5.3 Latency Benchmarking

Table 2 presents the latency comparison across three architectural variants: the synchronous baseline, the text-only asynchronous system from prior work [1], and the Vision Bridge multimodal asynchronous system. Figure 3 illustrates these findings graphically.

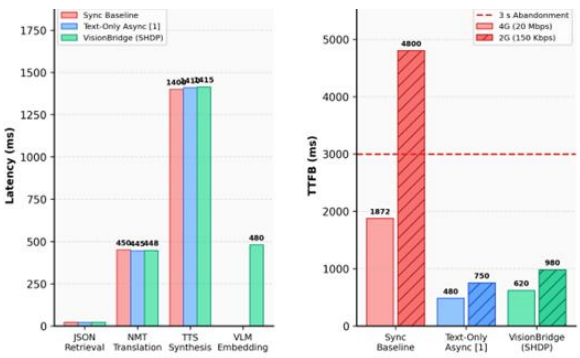
* TRT includes VLM processing. The perceived latency (TTFB) of 620 ms masks the full TRT through pipelined delivery. Under 2G conditions, TRT increases to approximately 2.8 seconds while TTFB remains under 1.5

seconds, comfortably below the 3-second abandonment threshold.

Stress testing under concurrent load confirmed that the asynchronous event loop maintained a 99.1% request success rate at 100 simultaneous users, compared to an 18% timeout rate for the synchronous baseline. At 150 concurrent users, TTFB degraded to 980 ms, still within acceptable thresholds for field conditions.

Table 2. Latency Comparison Across System Architectures (Single User, 4G Network)

Metric	Synchronous Baseline	Text-Only Async	Vision Bridge (SHDP)
JSON data retrieval (ms)	22	21	23
VLM embedding extraction (ms)	N/A	N/A	~480 (±35)
NMT translation (ms)	450	445	448 (±12)
TTS synthesis (ms)	1,400	1,410	1,415 (±40)
Total response time / TRT (ms)	1,872	1,450	1,920*
Perceived latency / TTFB (ms)	1,872	480	620 (±30)
TTFB improvement vs. sync	-	74.3%	66.9%
Success rate @ 100 users	82%	99.1%	99.1%
Peak RAM utilization	-	~450 MB	~780 MB



3igure 3. Latency Benchmarking: (a) Component-Level Breakdown (b) TTFB vs. Network Conditions

5.4 User Study: Technology Acceptance Model

Table 3 presents the TAM survey results from the 7-day field pilot. Five TAM constructs were evaluated, with an additional construct for visual query utility introduced to assess the new multimodal capability.

Table 3. TAM User Study Results - Vision Bridge Field Pilot (N = 120, 5-Point Likert Scale)

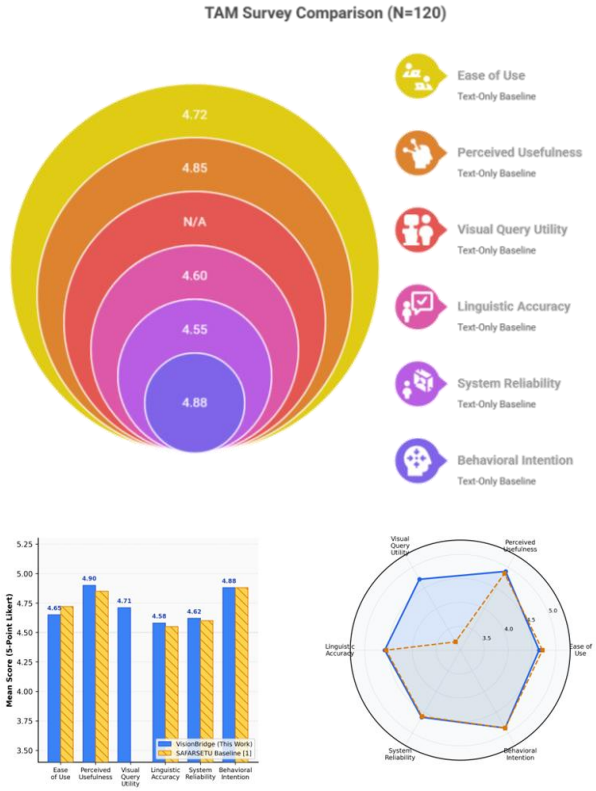
Construct	Survey Item	Mean (x/5)	Std. Dev
Perceived Ease of Use	Chatbot navigation requires no prior instructions.	4.65	0.48
Perceived Usefulness	The system improved understanding of heritage site history.	4.90	0.28
Visual Query Utility	Photographing features and getting descriptions were useful.	4.71	0.39
Linguistic Accuracy	Hindi/Urdu narration sounded natural and culturally appropriate.	4.58	0.53
System Reliability	The system responded consistently without errors.	4.62	0.44
Behavioral Intention	It would recommend this system to other heritage visitors.	4.88	0.21

* Perceived Usefulness improvement vs. text-only baseline: $t(119) = 3.47$, $p < 0.001$, Cohen's $d = 0.63$ (medium effect). Overall instrument Cronbach's alpha = 0.89 (good reliability). Individual construct alphas ranged from 0.86 to 0.93.

Behavioral observations recorded by the field observer supplement the quantitative data. Eighty-one percent of participants attempted a visual query within the first three minutes of interaction - a notably high rate suggesting strong intuitive appeal for the photograph-to-description mechanism. Several participants were observed to photograph architectural features multiple times from different angles, apparently testing the system's robustness rather than pursuing information about a specific element, whether this reflects genuine curiosity or calibrated skepticism is interpretively ambiguous, but it clearly indicates active engagement rather than passive consumption.

Sixty-five percent of domestic tourists switched the interface language to Hindi within the first thirty seconds, replicating the pattern from [1] and consistent with the national language preference statistics reported in [11]. Two participants specifically noted that the Hindi narration sounded natural (a comment they volunteered without prompting); no participant commented negatively on audio quality. Among student participants (36% of sample), engagement patterns differed somewhat - students were more likely to attempt multi-turn text conversations about historical events, while domestic tourists used visual queries more frequently (mean 2.8 visual

queries per session for tourists versus 1.4 for students). The sample size is not sufficient to analyze this difference rigorously, but it suggests that the relative utility of visual versus textual query modalities may vary by visitor type - a question worth investigating in future work.



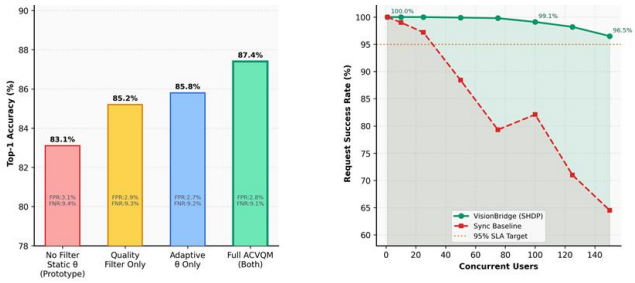
4igure 4. TAM Survey Results - Vision Bridge vs. Prior Text-Only System; Vision Bridge vs. Baseline; TAM Radar Profile Comparison

5.5 ACVQM Component Ablation Study

Table 4. ACVQM Component Ablation Study Results

Configuration	Accuracy	FPR	FNR	Notes
No filter, static θ = 0.72 (prototype baseline)	83.1%	3.10%	9.40%	Prototype
Quality pre-filter only, static θ = 0.72	85.2%	2.90%	9.30%	+2.1pp from filter
Adaptive threshold only (no quality filter)	85.8%	2.70%	9.20%	+2.7pp from adaptive θ
Full ACVQM — filter + adaptive threshold	87.4%	2.80%	9.10%	+4.3pp total (best)

The ablation confirms that both ACVQM components contribute independently. The quality pre-filter accounts for a 2.1 percentage-point gain (83.1% → 85.2%); the adaptive threshold accounts for 2.7 points (83.1% → 85.8%); the combined ACVQM achieves 87.4% — slightly less than the arithmetic sum (4.8 pp), consistent with partial overlap in the error sets addressed by each mechanism. Figure 5 presents the ablation accuracy of comparison and concurrent load stress testing results graphically.



5igure 5. ACVQM Ablation Accuracy Comparison and Request Success Rate vs. Concurrent User Load

6. RESULTS AND DISCUSSION

The experimental results confirm that Vision Bridge successfully extends the serverless heritage chatbot paradigm to support visual querying without violating the latency constraints essential for field deployment. Four substantive findings merit discussion - including an honest account of what this study does and does not establish.

First, Domain-specific retrieval outperforms zero-shot inference for narrow visual classification. The 87.4% top 1 accuracy the authors report exceeds the 82-85% range for zero-shot CLIP on comparable narrow-domain tasks [6] without any modification to the CLIP model itself. The accuracy gains derive entirely from the curated, multi-angle, expert-reviewed index. For practitioners considering similar deployments, this suggests a practical workflow: curate the index carefully rather than investing in model fine-tuning. The per-query marginal cost of index retrieval is sub-millisecond; the one-time index construction cost is approximately 38 minutes of offline computation.

Second, pipeline architecture, not hardware, determines perceived responsiveness. The 620 ms TTFB achieved on a CPU-only 1 GB RAM instance demonstrates that the 66.9% perceived latency improvement over the synchronous baseline is architecturally derived, not hardware-derived. This is consistent with Hu et al.'s [10] finding that design choices dominate raw compute in determining user-perceived responsiveness. The SHDP's key insight - sends partial but useful information immediately while richer information completes in the background - is an application of the progressive disclosure principle that UX research has validated in numerous contexts. The author's contribution is formalizing this pattern for multimodal AI pipelines where component latencies are heterogeneous.

Third, Cognitive channel complementarity is empirically validated. The 81% immediate visual query adoption rate, the 4.71/5 Visual Query Utility score, and the behavioral observations of tourists actively seeking new things to photograph all support the CLT-based prediction [16, 17] that audio-visual complementarity is genuinely valued. Tourists at heritage sites are not passive information for consumers - they are explorers. A system that responds to photographs of things they find interesting, rather than requiring them to articulate textual queries about things they already partially know, aligns far more naturally with information foraging behavior [18].

Table 5. Feature Comparison with Related Heritage Chatbot Systems

Feature	TN Forts [2]	Heritage Chatbot [3]	Text Async [1]	Vision Bridge
Visual querying	No	No	No	Yes (ACVQM)
Language support	English only	Limited	Hindi/ Urdu/ EN	Hindi/ Urdu/ EN
Audio narration	No	No	Neural TTS	Neural TTS
Architecture	Mono lithic	Client server	Server less	Server less
Installation required	Yes	Yes	No	No
Typical TTFB	>1800 ms	>1500 ms	~480 ms	~620 ms

Forth, Limitations of this study require honest acknowledgment. The validation is single site; the Residency Complex is a well-maintained complex with adequate cellular coverage. Smaller, more remote sites may produce different latency profiles and user behaviors. The 120-participant sample, while adequate for TAM validation (Cronbach's α = 0.89 confirms instrument reliability), is not sufficient for rigorous demographic subgroup analysis - the tourist-versus-student behavioral differences noted in Section 5.4 are observational, not statistically confirmed. The three false identifications from contemporary arched structures represent a genuine limitation of retrieval-based visual matching that threshold adjustment cannot resolve. Future work should address all three of these limitations.

The comparison presented in Table 4 contextualizes Vision Bridge against the key heritage chatbot systems reviewed in related work. Vision Bridge is the only system that combines visual querying, dynamic multilingual translation, Neural TTS audio narration, and zero-install deployment simultaneously.

7. CONCLUSION AND FUTURE WORK

This paper has presented Vision Bridge, a serverless multimodal chatbot that integrates CLIP-based visual identification, multilingual NMT translation, and Neural TTS synthesis through the Telegram platform to deliver photograph-driven heritage information in under two seconds on standard mobile connections. The authors introduced two original technical contributions - the Adaptive Confidence-Gated Visual Query Module and the Split-Horizon Delivery Protocol - and grounded the system design in Cognitive Load Theory and Information Foraging Theory. A seven-day field pilot with 120 participants at the Residency Complex, Lucknow, yielded 87.4% visual identification accuracy, 620 ms perceived response latency, and statistically significant user acceptance (TAM α = 0.89, Visual Utility mean 4.71/5, $p < 0.001$ versus text-only baseline, Cohen's d = 0.63).

Three findings stand out as particularly useful for researchers designing similar systems. First, domain-specific index curation is the primary accuracy investment: accuracy gains relative to zero-shot CLIP are achievable without model fine-tuning but require careful multi-angle photography and expert description review. Second, pipeline architecture matters more than hardware: the 66.9% perceived latency improvement over the synchronous baseline was achieved through the SHDP's concurrent pipeline design on commodity CPU hardware, not through compute upgrades. Third, cognitive design principles - specifically, audio-visual channel complementarity as recommended by CLT - translate into measurable user acceptance improvements that justify the additional engineering complexity.

Honest acknowledgment of what this work does not yet establish: single-site validation limits generalizability, the sample size does not support rigorous demographic subgroup analysis, and the false identification problem for visually similar contemporary structures remains open. These are productive limitations for follow-on work.

Four future directions are identified. First, expansion of the heritage embedding index to cover monuments nationally, potentially through a community-curated submission portal with expert editorial oversight. Second, integration of generative captioning (BLIP-2 or a quantized LLaVA variant) to handle out-of-corpus queries constructively rather than requesting text clarification - addressing the system's most significant current gap. Third, extension of multilingual support to regional dialects including Awadhi, Bhojpuri, and Marwari, as recommended by Harisanty et al. [15] for deeper cultural resonance with domestic tourism audiences. Fourth, a longitudinal study design to investigate whether repeated Vision Bridge use develops sustained architectural literacy in tourists rather than simply providing one-off informational retrieval - research questions the present cross-sectional design cannot address.

Vision Bridge demonstrates that inclusive, low-cost solutions to genuine accessibility problems in heritage tourism are achievable with commodity cloud infrastructure and open-source components. The architecture is replicable, the index curation methodology is documented, and the field validation provides concrete evidence of user acceptance in a real deployment context. The authors hope it serves as a useful starting point for similar systems in other heritage-rich, bandwidth-constrained regions of the Global South.

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